

How bats exit a crowded colony when relying on echolocation only – a modeling approach

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This **valuable** model-based study seeks to mimic bat echolocation behavior and flight under conditions of high interference, such as when large numbers of bats leave their roost together. Although some of the assumptions made in the model may be questioned, the simulations **convincingly** suggest that the problem of acoustic jamming in these situations may be less severe than previously thought. This finding will be of broad interest to scientists working in the fields of bat biology and collective behaviour.

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Abstract

Bats face a complex navigation challenge when emerging from densely populated roosts, where vast numbers take off at once in dark, confined spaces. Each bat must avoid collisions with walls and conspecifics while locating the exit, all amidst overlapping acoustic signals. This crowded environment creates the risk of acoustic jamming, in which the calls of neighboring bats interfere with echo detection, potentially obscuring vital information. Despite these challenges, bats navigate these conditions with remarkable success. Although bats have access to multiple sensory cues, here we focused on whether echolocation alone could provide sufficient information for orientation under such high-interference conditions. To explore whether and how they manage this challenge, we developed a sensorimotor model that mimics the bats' echolocation behavior under high-density conditions. Our model suggests that the problem of acoustic jamming may be less severe than previously assumed. Frequent calls with short inter-pulse intervals (IPI) increase the sensory input flow, allowing integration of echoic information across multiple calls. When combined with simple movement-guidance strategies—such as following walls and avoiding nearby obstacles—this accumulated information enables effective navigation in dense acoustic environments. Together, these findings demonstrate a plausible mechanism by which bats may overcome acoustic interference and underscore the role of signal redundancy in supporting robust echolocation-based navigation. Beyond advancing our understanding of bat behavior, they

also offer valuable insights for swarm robotics and collective movement in complex environments.

Introduction

In many bat species individuals dwell together in caves (or similar roosts), forming large colonies with tens to several millions of individuals^{1,2}. Each evening, at approximately the same time, the bats take off from their roost, navigating through its passages toward the exit. The high density of bats flying simultaneously in great proximity poses many challenges for orientation in such a crowded and noisy environment. Flying while avoiding collisions, often in a pitch-black cave, demands the continuous detection and localization of both obstacles and nearby bats^{3,4}. Employing echolocation, bats emit strong ultrasonic signals and interpret the reflected echoes to perceive their surroundings⁵. The reception of neighbors' loud calls, which share similar acoustic features with their own calls, can potentially hinder the bats' ability to detect the faint echoes reflected off the walls and the surrounding bats^{5,6}. We examined whether bats could rely solely on echolocation to exit the roost even during such a chaotic 'rush hour'.

The question of how bats cope with acoustic interference — i.e., the masking of potential echoes by conspecific signals — has been extensively researched using playback experiments, field observations, on-body tags, and computational simulations^{7–17}. However, much of this research has focused on foraging bats in small groups^{5,6,9,16,18–20}. The challenges bats encounter during roost exits (e.g., cave exits) differ markedly from those encountered during group foraging. Bat density during roost exits is significantly higher, and bats need to detect and follow static walls or obstacles, which produce loud echoes, rather than small, sporadic prey items that generate faint echoes²¹. Their flight during exits is also more directional and involves avoiding collisions with conspecifics, in contrast to the erratic hunting maneuvers typically observed while foraging. Echolocation studies during dense collective movement are scarce^{4,6,22–25}, likely due to the complexities in recording separate echolocation calls and tracking individual flights within the swarm.

While collective movement has been extensively studied in various species, such as insect swarming, fish schooling, and bird murmuration^{26–32}, as well as in swarm robotics, where agents perform tasks such as coordinated navigation and maze-solving^{33–35}, most studies have focused on movement algorithms that assume full detection of neighbors^{36–43}. Some models have incorporated limited interaction rules where individuals respond to only one or a few neighbors due to sensory constraints^{44,45}. However, fewer studies have explicitly examined how sensory interference, occlusion, and noise influence decision-making and affect collective movement⁴⁶.

The present study addresses these gaps by introducing an agent-based sensorimotor model based on the well-documented echolocation capabilities of bats, simulating multiple bats pathfinding their way out of a cave-like structure. We modeled the echolocation behavior of two insectivorous bat species: *Pipistrellus kuhlii* (PK), which roosts in abandoned buildings and frequently navigates through conspecific-dense, cluttered corridors and the cave dwelling *Rhinopoma microphyllum* (RM) which emerges from its roosts with thousands of individuals simultaneously. These two species differ in their echolocation signals – PK echolocation signals are characterized by a wider bandwidth and a higher terminal frequency than RM calls. We quantified the performance of an individual bat flying among conspecifics, demonstrating that even a relatively simple sensorimotor algorithm can facilitate successful orientation in such complex environments. The modeling approach enabled us to explore how various biological and ecological factors influence successful navigation under such challenging conditions.

Results

Our model was designed with conservative assumptions regarding bats' sensing, movement, and sensorimotor integration, aiming to underestimate their capabilities and thereby establish a lower bound on their actual performance. Real bats likely outperform the model's predictions. In our 2D simulations^{7,47}, each bat emits sound signals and receives echoes reflected from the roost walls and other bats, while also encountering masking signals caused by calls from conspecifics. These masking signals can interfere and completely eliminate echo detection (which we refer to as jamming) or cause echo localization errors. After estimating the distance and direction of each detected reflector, the bat adjusts its echolocation parameters and maneuvers to find the exit while simultaneously avoiding collisions. The bats dynamically adjust their echolocation parameters—including call rate, duration, and frequency range—based on the estimated distance to obstacles, following the well-documented transition between search, approach, and buzz phases observed in echolocating bats (see^{7,47} and Methods). Their reception was modeled using a biologically inspired filter-bank receiver comprising 80 gammatone channels^{7,47,48}. Each bat adjusted its flight following a simple pathfinding algorithm based solely on the estimated locations of the detected reflectors (see Methods, Supplementary Figure 1, and Supplementary Movie 1 for additional details). The bats had to exit a roost designed as a corridor (14.5 m long x 2.5 m wide), with a right-angle turn located 5.5 m before the exit (**Figure 1A**⁴⁷). Additionally, an obstacle (1.25 m wide) was situated 2.25 m in front of the opening. The simulated bats initiated their flight from the far end of the corridor, within a randomly selected $1.5 \times 2 \text{ m}^2$ area, taking off in the general direction of the exit (± 30 degrees), without prior knowledge of the roost's structure.

The sensory model accounted for six types of acoustic signals: (1) the bat's own calls, (2) echoes from conspecifics, (3) echoes from walls in response to the bat's own calls (i.e., desired wall echoes), (4) echoes from conspecific calls reflected off other bats, (5) echoes from conspecific calls reflected off walls, and (6) the conspecific calls themselves. In the baseline model, bats were assumed to reliably distinguish between all these signal types. In contrast, the confusion model described below specifically tested the impact of failing to distinguish between desired wall echoes (3) and wall echoes generated by conspecific calls (5), while preserving the bat's ability to identify all other signal types. In brief, the bat responded to echoes as follows (see Methods and Supplementary Figure 1 for details): If an obstacle or a conspecific was detected in front of the bat and was too close, the bat would maneuver to avoid a collision. Otherwise, for exit-seeking, the bat would follow the contour of the walls by steering toward the farthest detected obstacle ahead. If a gap greater than 0.5 m was identified between adjacent reflectors, the bat directed its trajectory toward the center of the gap.

The ability of the bats to exit the roost within 15 sec was evaluated for different group sizes, from a single bat and up to 100 individuals. For simplicity, we will refer to the initial density at the cave's far end as the number of bats per 3m^2 (i.e., for groups of 100 bats, the density is 100 bats/ 3m^2 , or 33.3 bats/ m^2). The bat densities we tested were chosen to reflect the typical range of bat densities observed in natural caves during emergence events^{25,49,50}. Key model parameters, such as the sensory integration window, object target strength, echolocation parameters, and flight velocity (see **Table 1**⁴⁷), were manipulated and their impact on the exit performance was analyzed. To explicitly quantify the effect of sensory masking vs. the effect of collision avoidance (i.e., spatial interference) only, we turned the acoustic interference on and off to measure its impact. The number of repetitions for each scenario was as follows: 1 bat: 240; 2 bats: 120; 5 bats: 48; 10 bats: 24; 20 bats: 12; 40 bats: 12; 100 bats: 6 (see **Table 1**⁴⁷).

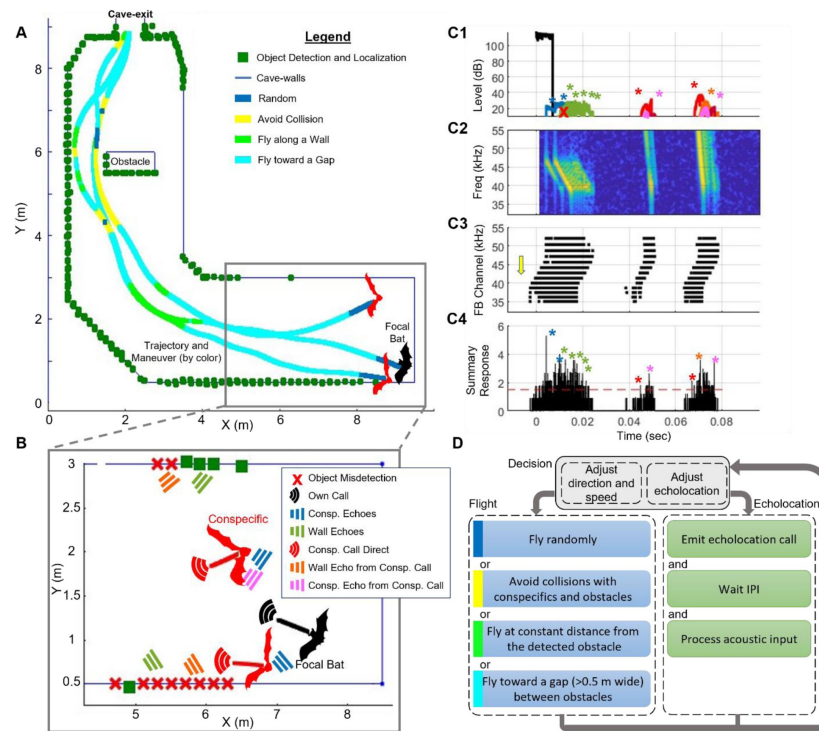


Figure 1

The sensorimotor model.

(A) Figure 2: The sensorimotor model. (A) Top view of the cave with three bats' trajectories. The focal bat is shown in black. All bats' flight trajectories are displayed while the bats' moment-to-moment decisions are represented by the colored lines: blue – random flight, yellow – collision avoidance, light green – wall-following, turquoise – movement toward a wall gap (see panel D for details). Green squares depict reflectors detected by the focal bat along its route. **(B)** A zoomed-in view of the marked rectangular area in Panel A, where the focal bat (black) emitted one echolocation call (black) and received echoes from the cave walls (green) and from two other bats (blue). It also received conspecifics' calls (red) and their reflection from the cave walls (orange), as well as the reflections from other bats (pink). Green squares indicate points that were detected by the focal bat from this call and red x's indicate missed points due to acoustic masking (i.e., jammed reflectors). The locations of the detected reflectors (green squares) are marked according to their localization by the bat (with simulated errors). The lines near the bats depict their flight direction. **(C)** The acoustic scene received by the focal bat is as depicted in B, including the emitted call and all received signals (colors as in panel B). **(C1)** The time-domain plot displays the envelope of signals, encompassing the emitted call and the received signals: the desired echoes from the walls and conspecifics; the calls of other bats; the echoes returning from conspecific calls and reflected off the walls and off other bats. Notably, in this example, some of the desired wall-echoes are jammed by stronger self-echoes reflected from nearby conspecifics. **(C2)** The spectrogram of all the received signals presented in C1; for clarity, the emitted call is not depicted. **(C3)** The responses of the active channels of the cochlear filter bank (FB channel) after de-chirping. Each channel is represented by its central frequency on the y-axis. Each black dot represents the timing of a reaction that was above the detection threshold in each channel. Note that early reactions in low-frequency channels (marked by a yellow arrow) result from the stimulation of those channels caused by the higher frequencies of the downward FM chirp. However, most of these stimulations do not reach the detection threshold and are therefore not detected (see Methods). **(C4)** The detections of each channel are convolved with a Gaussian kernel, summed, and compared with the detection threshold (dotted red line). Colored asterisks mark peaks that were classified as **successful detections**—those identified in both the interference-free and full detection conditions (see Methods for details). Other peaks may originate from masking signals or overlapping echoes that did not meet the detection criteria (colors of the sources are as defined above). **Panel D** depicts the pathfinding algorithm used by the bat. The algorithm involves a correlated-random flight during the search phase (blue), collision avoidance (yellow), flying along the wall at a constant distance (green), and flying toward the center of a gap between obstacles as an indicator of a possible exit (cyan). After each echolocation call, the bat awaits an IPI (Inter Pulse Interval) period before processing the detections, adjusting flight and echolocation parameters, and emitting the next call. Based on the received signals, it then modifies its next call design and adjusts its direction and speed accordingly. For a detailed diagram of the complete sensorimotor process see Supplementary Figure 1.

Key parameter	Tested range	Default value	Effect Size on Explained Variable				GLM Explaining factors
			Exit prob. (%)	Time-to-exit (sec)	Jamming-prob. (%)	Obs. Collision (sec ⁻¹)	
Number Of Bats	[1, 2, 5, 10, 20, 40, 100]	All Values	-0.37/bat* [63:100]	0.044/bat * [4.6:8.9]	0.54/bat* [0:54]	0.25/bat* [0.05:0.3]	Number of bats
Bat species	[PK, RM]	PK	4.5 [63:67]	-0.4* [8.5:8.9]	9* [54:63]	0.01 [0.29:0.3]	Number of bats, bat species
Integration Window (#)	[0,1,3,5,10]	5	0.41/call* [29:70]	-0.16/call* [9.2:7.6]	0.01/call [54:55]	0.27/call [0.25:0.53]	Number of bats, integration window size
Nominal flight speed (m/s)	[2,4,6,8,10]	6	6/(1m/s), -12/(1m/s)* [15:63]	-1/(1m/s), 1/(1m/s)* [6:9.6]	1.4/(1m/s)* [55:65]	0.18/(1m/s)* [0.07:1.3]	Number of bats, flight speed, square of flight speed
Call level (dB-SPL, @ 0.1m)	[100,110,120, 130]	120	13 ^ψ [50:63]	-0.5 [7.9:8.4]	0.5/10dB* [53:58]	0.07 ^ψ [0.29:0.36]	Number of bats, call level
Misidentification	[Yes/No]	N	-69* [14:83]	1.3* [7.6:8.9]	30* [50:80]	0.6* [0.2:0.8]	Number of bats, confusion flag
Misidentification and multi-call clustering	[Yes/No]	N	-23* [58:83]	1.7* [7.6:9.3]	29* [50:79]	-0.02* [0.18:0.2]	Number of bats, process flag
Masking	[Yes/No]	Y	23* [63:86]	0.8* [8.0:8.8]	54* [0:54]	0.03 [0.26:0.29]	Number of bats, masking
Wall target strength (dB) ^α	[-33, -23, -13, -3]	-23	16/10dB * [23:87]	0.7/10dB * [6.6:9.5]	-8.5/10dB * [34:68]	0.07/10dB * [0.19:0.47]	Number of bats, wall target strength
Conspecific target strength (dB) ^β	[-49, -43, -33, -23]	-23	-1.5/10dB [85:91]	0.25/10dB* [7.2:8.15]	-0.5/10dB [48:50]	0.1/10dB [0.16:0.2]	Number of bats, conspecific target strength

Table 1

Key model parameters and their effects on performance metrics.

The table presents the key parameters tested, their ranges, default values, and effect sizes on various performance metrics: exit probability, time-to-exit, jamming probability, and collision rate with obstacles. The parameters comprised the number of bats, bat species (PK-*Pipistrellus kuhlii*, RM -*Rhinopoma microphyllum*), integration window, nominal flight speed, call level, echo mis-identification with multi-call clustering (yes/no), masking (yes/no), wall target strength, and conspecific target strength. In each scenario, all parameters except the tested one were set to the default value. Call levels are reported in dB-SPL, referenced at 0.1 m from the source. Effect sizes for each parameter are explicitly listed for all four-performance metrics, expressed as the change per unit of the tested parameter (e.g., per bat or per 10 dB). For flight speed, a non-monotonic relationship was observed, and values are reported both before and after the peak performance (see Results, [Fig. 3B](#)). Square brackets present the minimum and maximum values of the metric across the tested range, with the maximal effect size in 100 or 40 bats densities. Asterisk (*) indicates a significant impact. Each scenario was tested using Generalized Linear Models (GLMs) with number-of-bats and the tested parameters set as fixed explaining variables. Exit probability and jamming probability were treated as binomially distributed, collision rate was treated as a Poisson distributed, and all other variables were considered normally distributed. Explaining variables were set as fixed factors. The number of repetitions for each scenario was as follows: 1 bat: 240; 2 bats: 120, 5 bats: 48; 10 bats: 24; 20 bats: 12; 40 bats: 12; 100 bats: 6. ^ψ A significant difference in call intensity was found only for a bat density of 100 bats/3m², and between the group with a level of 100dB-SPL and all other groups. ^α see Supplementary Figure 3. ^β see Supplementary Figure 4.

Bats find their way out of the cave even at high conspecific densities

We first examined how bat density affects bats' ability to exit the cave, both alone and in a group. The probability of exiting the cave within 15 seconds—defined as the proportion of bats that successfully exited within this time frame—was significantly reduced at higher densities (**Figure 3A**, see Supplementary Movie 1 for a view of the bats' movement, $p < 10^{-10}$, $t = -23$, $DF = 4077$, GLM, see details in **Table 1**). In trials in which a single bat was flying alone, it successfully exited the cave in 100% of the cases. Even without sensory interference, the probability of exiting decreased significantly from 100% to $86\% \pm 1.4\%$ and $91\% \pm 1.7\%$ at densities of 100 PKs/ 3m^2 and 100 RMs/ 3m^2 , respectively (mean $\pm$ s.e.). When acoustic interference was added, the exit probability further decreased to $63\% \pm 1.4\%$ and $67\% \pm 1.4\%$ for 100 PKs and RMs, respectively (see **Figure 3A**).

The difference in exit probability between the two species was not significant ($p = 0.08$, $t = 1.74$, $DF = 4077$, GLM as above, **Figure 3A**). Similarly, the difference in echolocation parameters between the two species did not affect the collision rate with the walls (with a maximum of 0.29 and 0.3 collisions per bat per second for PK and RM, respectively, with 100 bats ($p = 0.63$, $t = -0.48$, $DF = 4077$, GLM, **Figure 3C**, see details in **Table 1**). To quantify sensory interference, we defined a jammed echo as an echo entirely missed due to masking. The jamming probability, which was calculated as the number of jammed echoes divided by the total number of self-echoes, was significantly higher for RM compared to PK. The maximum difference between the two models was 14.3% at a density of 10 bats, with a smaller difference of 9.8% observed at 100 bats ($p < 10^{-10}$, $t = 6.56$, $DF = 4077$, GLM, **Figure 3D**, see details in **Table 1**). Accordingly, PK demonstrated a minor but significant advantage in detecting the cave walls ($p = 0.024$, $t = -2.25$, $DF = 4077$, GLM, **Figure 3E**, see details in **Table 1**). With 100 bats flying together, the probability of detecting a wall echo at a distance of 1 m in a single call was around 50% and 46% for PK and RM, respectively. Despite this minor disadvantage in detection, RM bats exhibited a better time-to-exit average than PK bats, being 0.5 seconds faster to exit ($p = 0.0005$, $t = -4.06$, $DF = 3533$, for $n = 40$ bats, **Figure 3B**). Additionally, RM bats experienced a significantly higher probability of their self-generated echoes, reflected off conspecifics, being jammed ($p = 0.00016$, $t = 3.8$, $DF = 3593$, GLM; see details in **Table 1**).

Multi-call integration improves exit performance

We next examined whether bats improve their performance when integrating information from several consecutive calls. The integration window determines the number of previous calls the bat uses at each step to guide its next movement decision (see Methods and Supplementary Figure 2A). In the basic multi-call integration model, detections from the previous calls — by default the last five — were stored in an allocentric (x-y) reference frame, with each detection treated independently as a potential obstacle without clustering or filtering. At each decision, the bat takes all of these detections into account when guiding its movement and echolocation. The probability of exiting the roost significantly increased when increasing the size of the integration window for all bat densities ($p < 10^{-10}$, $t = 28.5$, $DF = 10197$, GLM, **Figure 4A**, see details in **Table 1**). For example, at a density of 40 bats/ 3m^2 , the exit probability improved from 20%, to 75%, and to 87% as the window size increased from one, to three, and to 10 previous calls, respectively. In addition, increasing the window size resulted in a significant improvement in the time-to-exit and the avoidance of wall collisions ($p < 10^{-10}$, $t = -12.8$, $DF = 7661$; $p < 10^{-10}$, $t = -46.5$, $DF = 10197$, respectively, GLM, see details in **Table 1**). With 100 bats, the collision rate decreased by a factor of 2 from 0.53 to 0.25 collisions per second as the window increased from 1 to 10 calls. The size of the integration window had no significant effect on the jamming probability ($p = 0.37$, $t = 0.9$, $DF = 10197$, GLM, see details in **Table 1**).

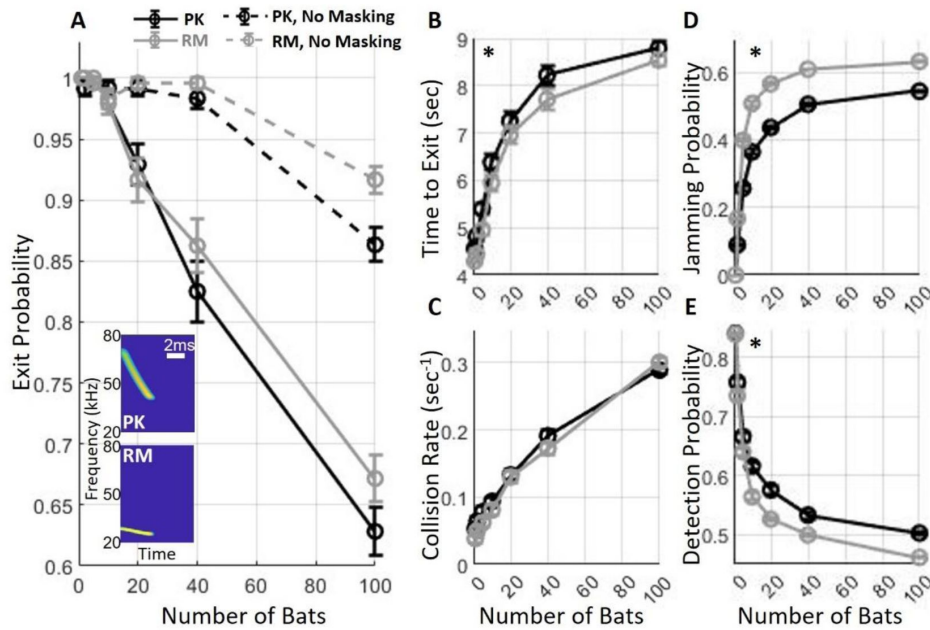


Figure 3

Exit performance of *P. Kuhlmanni* (PK) and *R. Microphyllus* (RM).

((A) Sensory interference significantly impaired the probability of exiting the cave (compare dashed lines with continuous lines). The probability of a successful exit also declined as the number of bats increased, with no significant difference observed between the species when masking interference was applied. The insert shows the spectrograms of the echolocation calls of PK (top) and RM (bottom). (B) The time-to-exit, which was calculated for successful trials only, and (C) the collision rate with the walls both increased as a function of the number of bats. (D) The probability of jamming significantly increased to about 55% and 63% with 100 bats for PK and RM, respectively. (E) The detection probability of a wall reflector at one meter or less in front of a bat decreased as a function of the number of bats. In panels (A-E), circles represent means and bars represent standard errors (see details in [Table 1](#)). Asterisks indicate significant differences between the lines in each panel.

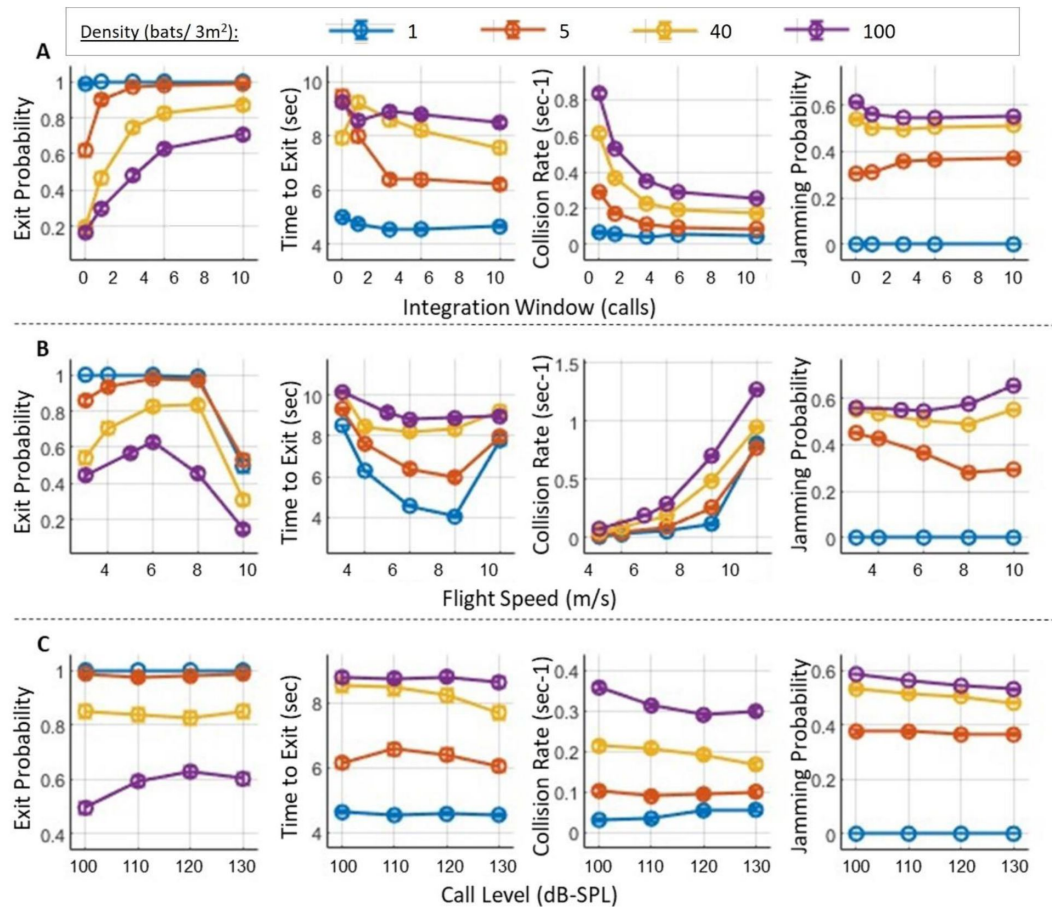


Figure 4

Exit performance as a function of key sensorimotor parameters.

(A) The effect of the integration window on the probability of exiting the roost, the time-to exit, the rate of collisions with the walls, and the probability of jamming (from left to right, respectively). Each colored line shows the trend as a function of the window-size for different bat densities, with each color representing a specific density. Note that a window size of 0 indicates that only the most recent call is used in the bat's decision-making, without integrating detections from previous calls. **(B)** The effect of the nominal flight speed of the bats, with panels and line-colors as in panel A. An optimal speed of approximately 6 to 8 m/sec can be observed for all densities above one bat. **(C)** The effect of call intensity on exit performance, panels as in (A). In all panels, circles represent means and bars represent standard errors. Error bars depicting standard errors are presented but are very small due to the large number of simulation repetitions. See [Table 1](#) for the number of simulated bats.

Exit probability was maximal at an intermediate flight-speed

We observed a significant and non-linear effect of the flight speed of the bats on the performance, as shown in **Figure 3B** ($p < 10^{-10}$, $t = -29.9$, $DF = 10196$, GLM, see details in **Table 1**). The exit probability increased with flight speed until it reached a maximum at 6-8 m/s and then declined rapidly. This was the case for all bat densities, with the maximal exit probability ranging between 65% to 99%. At the optimal velocity, the time-to-exit was also minimal. However, the collision rate increased monotonically with speed, with a steep incline above the optimal speed.

Call intensity had only a minor effect on exit performance and only at high bat densities

For low bat densities (< 40 bats), call intensity did not have a significant impact on either exit probability or collision rate (**Figure 4C**, $p = 0.89$, $t = 0.13$, $DF = 5757$; $p = 82$, $t = -0.21$, $DF = 5757$, respectively, GLM, see details in **Table 1**). Call intensity affected exit performance only when the intensity dropped to 100 dB-SPL (@ 0.1m) and only at a high bat density of 100 bats/ $3m^2$ (**Figure 4C**). In this scenario, the exit probability declined from approximately 60% to 49.5% ($p = 0.003$, $F = 8.45$, $DF = 2396$, One-way ANOVA with 'hsd' post hoc test), and the collision rate increased from 0.3 to 0.35 collisions per second ($p < 3 \cdot 10^{-6}$, $F = 22.18$, $DF = 2396$). Notably, this low intensity is below the typical search-call intensity of most echolocating bats. At the same bat density (100 bats/ $3m^2$), further increasing the call intensity to above 100dB-SPL had no significant effect on either exit probability ($p = 0.6$) or collision rate ($p = 0.07$). Calling louder also slightly, but significantly, decreased the jamming probability at all bat densities, with a decrease of $3.5\% \pm 8\%$ to $5.5\% \pm 5\%$ (mean $\pm$ s.e.) ($p = 0.02$, $t = -2.26$, $DF = 8157$, GLM, see **Table 1**).

While confusion between the desired echoes and those from conspecific calls may significantly impair exit performance, multi-call clustering helps to mitigate this

We next addressed the challenge of echo classification, assuming that a bat can differentiate an echo resulting from its own calls from echoes resulting from the calls of other bats. To examine this assumption, we tested another model, referred to as the **confusion model**, in which bats responded similarly both to wall echoes returning from their own emissions and to those from conspecific emissions, treating all as their own echoes. This confusion significantly decreased exit performance for all bat densities (above one bat). The probability of a successful exit for a density of 40 bats/ $3m^2$ dropped from $83.3 \pm 2.4\%$ to $14.6 \pm 2.3\%$ ($p < 10^{-10}$, $t = -20.7$, $DF = 2877$, GLM, see details in **Table 1**), the exit time increased from 7.6 ± 0.18 to 9.3 ± 0.2 seconds ($p < 10^{-10}$, $t = 15.5$, $DF = 2157$, GLM), and the collision rate increased significantly from 0.2 ± 0.007 to 0.8 ± 0.013 collisions per second ($p < 10^{-10}$, $t = -30$, $DF = 28777$, GLM, see **Figure 5**, red and yellow lines).

To further examine whether this substantial decrease in performance could be mitigated even without improving echo identification, we tested an enhanced integration model that, in addition to extending the number of calls integrated, clustered spatially close detections, removed outliers, and estimated wall directions based on grouped reflectors (see Methods and Supplementary Figure 2B). This **'multi-call clustering'** significantly improved performance, but exit probability and time-to-exit still remained significantly lower than without echo-confusion: exit probability = $58 \pm 3\%$ in comparison to $83.3 \pm 2.4\%$ without echo confusion ($p < 10^{-10}$, $t = 18.3$, $DF = 28777$, GLM), time-to-exit = 9.3 ± 0.2 seconds ($p < 10^{-10}$, $t = -13.7$, $DF = 1996$, GLM), see **Figure 4**, yellow line. The results above are reported for a density of 40 bats/ $3m^2$. Interestingly, the multi-call clustering restored the collision rate to the levels observed under the "No Confusion" condition ($p = 0.68$, $t = -0.42$, $DF = 2877$, GLM, see **Figure 5C**, dark-purple and red lines).

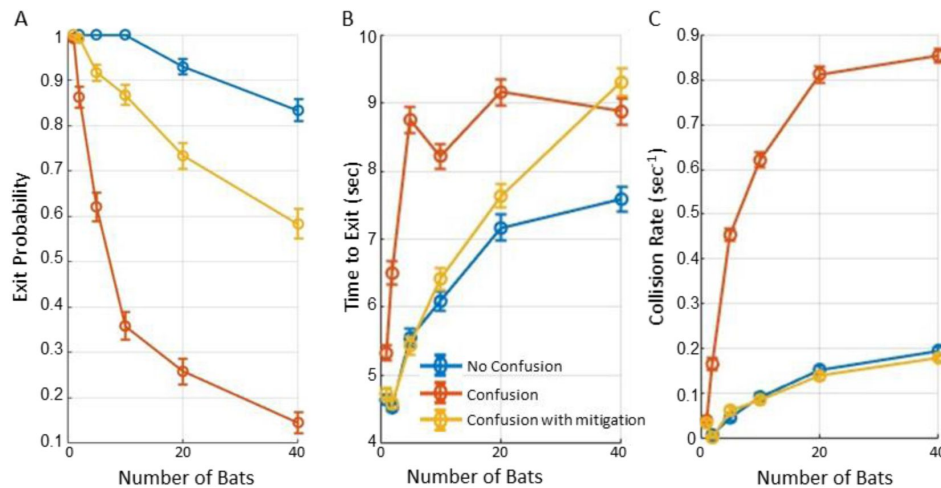


Figure 5

The impact of confusion on performance.

The figure illustrates the impact of classification confusion on roost-exit performance under various conditions. Blue lines depict trials with masking, while assuming that bats can distinguish between echoes from their own calls and those of conspecifics (referred to as “No Confusion”). Red lines depict performance where confusion between echoes is assumed. Yellow lines depict performance under the confusion condition, with the added capability of multi-call clustering in a short-term working memory (referred to as “confusion with mitigation”, see text for further details). In all panels, circles represent means and bars indicate standard errors. **(A)** The probability of exiting the roost significantly decreased with masking and confusion. In conditions with confusion and no aggregation process, only 15% of bats successfully exited the roost, at a density of 40 bats/3m². Multi-call clustering partially mitigated the confusion effect but did not eliminate it. **(B)** Bats with the ability to distinguish between echoes demonstrated significantly shorter exit times than those experiencing confusion. Note that time-to-exit refers only to successful attempts. **(C)** The collision rate with walls was highest for bats experiencing both masking and confusion but decreased significantly when without confusion. Multi-call clustering restored performance to the “No Confusion” condition, reducing collision rates accordingly, at densities between 1 to 40 bats/3m².

Effect of Wall and Conspecific Target Strengths on Exit Performance

Increasing the wall target strength significantly enhanced navigation performance (Supplementary Figure 3, [Table 1](#)), improving exit probability by up to 64% and reducing time-to-exit by up to 2.8 seconds ($p < 10^{-10}$). Stronger wall echoes improved environmental awareness but also slightly increased masking of desired conspecific signals.

In contrast, changes in conspecific target strength had a much smaller effect (Supplementary Figure 4, [Table 1](#)), with only minor improvements in detection and collision rates, and no significant impact on exit probability. This likely reflects the fact that both desired and masking signals scale similarly with conspecific reflectivity. Overall, the model showed low sensitivity to variations in conspecific target strength.

Discussion

We present a model-based approach that suggests how echolocating bats might find their way out of a crowded roost while contending with severe sensory interference caused by numerous nearby conspecifics. Our results demonstrate that a single bat, lacking prior knowledge of the roost's structure, successfully found the exit in all simulated trials using echolocation alone. As bat density increases, the bats face increased collision risks and more substantial acoustic interference, both of which reduce the probability of efficiently finding the exit. Nevertheless, even at densities of 100 bats/3m², most bats (63%) successfully exited the roost within a short timeframe. These results are based on a 2D simulation with up to 33 bats/m², under the assumption that bats can distinguish their own echoes from those of conspecifics. We demonstrate how a simple sensorimotor approach can solve this supposedly challenging task. This approach encompasses the following principles: (1) emission of echolocation calls; (2) reception of reflected echoes and masking signals; (3) detection of reflectors (including walls and conspecifics) using a gammatone filter bank biological receiver; (4) localization of the detected objects; (5) employment of multi-call integration of acoustic detections; (6) adjustment of flight and echolocation behavior based on the distance and angle to the reflectors; and (7) application of simple pathfinding rules to follow walls and gaps while avoiding collisions. Notably, despite the jamming of a substantial percentage of the echoes — particularly, with 100 bats, 50% of the echoes from nearby obstacles at ~1 m distance — the bats managed to maneuver correctly even with this simple approach and partial data.

A key component of this success was the multi-call integration: increasing the number of stored calls from one to ten markedly improved performance, raising the exit probability from 20% to 87% and halving the collision rate. Real bats likely use a much more sophisticated approach that also includes memorizing the roost's structure⁵¹, using landmarks inside the roost⁵², reliance on the movement of nearby conspecifics^{43,49}, and exploitation of other sensory modalities. We thus expect their actual performance to surpass that of our modeled bats.

Our model suggests that acoustic jamming might be less problematic than has been generally assumed^{5,11,53}, and that movement under severe acoustic masking could be mitigated by increasing the call-rate, creating a redundancy across several calls-similar to how real bats behave in a complex environment⁶. In our model, the Inter-Pulse Interval (IPI) naturally varied according to established echolocation behavior, decreasing from 100 msec in the search phase to 35 msec (~28 calls per second) in the approach phase, and further to 5 msec (200 calls per second) during the final buzz ([Table 2](#)). The results indicate that this redundancy, combined with simple sensorimotor heuristics, enhances successful navigation. This is consistent with several recent studies that have pointed in this direction^{7,24,25}.

<i>Pipistrellus kuhlii</i> (Kuhl's pipistrelle)						
Flight phase	Search	Approach		Buzz		
Parameter		Start	End	Terminal 1 start	Terminal 1 end	Terminal 2
Inter Pulse Interval [ms]	100	70	35	18	6	5
Call duration [ms]	7	5	2	2	1	0.3
Terminal frequency [kHz]	39	39	39	39	39	39
Chirp bandwidth [kHz]	8	35	30	30	20	20
Call intensity [dB-SPL]	120	120	90	90	80	80
Distance to target [m]	>1.2	1.2	0.4	0.4	0.2	<0.2
<i>Rhinopoma microphyllum</i> (greater mouse-tailed bat)						
Flight phase	Search	Approach		Buzz		
Parameter		Start	End	Terminal 1 start	Terminal 1 end	Terminal 2
Inter Pulse Interval [ms]	100	80	20	18	10	9
Call duration [ms]	12	7	2	2	1.5	0.75
Terminal frequency [kHz]	26	26	26	26	26	23.5
Chirp bandwidth [kHz]	3	4	5	3	3	3
Call intensity [dB-SPL, @0.1m]	120	120	90	90	80	80
Distance to target [m]	>1.2	1.2	0.4	0.4	0.2	<0.2

Table 2

Echolocation parameters.

The table presents the echolocation parameters of the two bat species we simulated during the specified flight phases (i.e., search, approach, buzz, and final buzz). In each phase, except for the search phase, in which the parameters remain constant, the parameters for each call are determined by the distance to the closest detected object.

The bat densities we simulated, ranging from 1 to 100 bats per 3m², reflect a wide range reported in field studies. Although bat colonies can be much larger than 100 bats, the maximal simulated density in our model (100 bats per 3 m²) resulted in bats flying in very close proximity, with an average nearest-neighbor distance of 0.27 meters. This density is higher than some of the most-dense reported bat aggregations, including studies on *Miniopterus fuliginosus*⁴⁹, *Myotis grisescens*⁵⁴, and *Tadarida brasiliensis*^{4,50,55}, where bats emerge from the roost at rates of 15 to 500 bats per second, but fly with an average distance of 0.5 meters between individual bats.

We compared the performance of two FM echolocating insectivorous bat species: *Pipistrellus kuhlii* (PK) and *Rhinopoma microphyllum* (RM). PK bats emit wideband echolocation signals that are less prone to jamming than RM bats' narrowband signal^{15,56}, as wideband signals distribute energy across a broader frequency range and are thus more robust against interference^{9,24,57}. Our findings show that PK signals slightly reduce jamming probability (by 9%) and improve wall detection. However, no significant differences in exit probabilities were noted between the two species.

Using a simulation allowed us to separate the effects of **acoustic interference (masking)** and **spatial interference (collision avoidance)** and revealed new insights into the sensorimotor strategy that that could plausibly be used by real bats. The spatial interference reduced the probability of exiting the roost from 100% to 87%, while the acoustic masking further decreased it to 63%. Increasing call intensity had little effect on exit performance, although slightly improving it at high bat densities. When all bats increased their calling intensity, both desired echoes and masking signals intensified equally, resulting in only a marginal effect. This was tested by varying call intensity levels (100-130 dB SPL) in our simulations (**Table 1**), demonstrating that beyond a certain level (~110 dB SPL), there is no further benefit in improving obstacle detection. These results align with previous studies that have drawn similar conclusions^{7,24}.

Bats constantly adjust their flight speed to their surroundings^{58–61} and specifically when conspecifics are nearby⁶². Our study suggests that the optimal velocity for flying through a crowded roost ranges from 6 m/sec to 8 m/sec for densities of 2-100 bats/3m². Exceeding this velocity-range led to a significant drop in exit probability due to a significant increase in wall collisions. We found that this speed did not depend on bat density in accordance with the observations of Theriault et al.⁵⁰. Notably, the reported velocities of RM when exiting a cave²⁵ and PK emergence velocity near the cave⁶³ are close to the speed that appears optimal, based on our simulations.

We also tested the effects of wall and conspecific target strengths on navigation. Stronger wall echoes substantially improved exit probability and reduced obstacle collisions, despite slightly increasing masking of conspecific echoes (Supplementary Figure 3). In contrast, changes in conspecific reflectivity had minimal impact, likely because both desired and masking signals scaled similarly (Supplementary Figure 4). This result may also stem from our model's assumption that bats slow down, but continue flying at the same direction following a collision with a conspecific.

Our basic model assumed that bats can distinguish between wall echoes and conspecific echoes, as demonstrated in previous studies^{64–66}. We suggest that this is a feasible assumption because echoes from cave walls are longer and exhibit distinct spectro-temporal patterns, whereas echoes from smaller objects, such as conspecifics, are shorter^{47,67,68}. However, wall echoes reflected from conspecific calls might resemble those from the bat's own calls in their amplitude and time-frequency characteristics^{20,62,69}. This led us to question how the misidentification of such echoes as obstacles might affect navigation. When unable to distinguish between these echoes, the simulated bats responded to all as if they were their own and thus mis-localized conspecific wall echoes. The confusion led to a substantial drop in exit performance, with only 15% of the bats successfully exiting compared to 82% under no-confusion conditions, at a density

of 40 bats/3m². At the same time, the collision rate increased markedly from 0.2 to 0.85 collisions per second. These results demonstrate the vital importance of echo discrimination for successful navigation, highlighting both the necessity of distinguishing between self and conspecific echoes and the classic challenge of detecting desired signals in noisy environments. There is a substantial evidence in the literature supporting the assumption that bats can recognize their own echoes and reliably distinguish them from those of conspecifics^{57,64–66,70}.

Previous studies have also demonstrated that bats can aggregate acoustic information received sequentially over several echolocation calls, effectively constructing an auditory scene in complex environments^{5,71–75}. Bats are also known to emit call sequences in groups, particularly when spatiotemporal localization demands are high. Studies have recorded sequences of 2-15 grouped calls, supporting the idea that grouping facilitates echo aggregation^{72,76}. Accordingly, we tested how multi-call clustering process—which included grouping nearby reflectors, removing outliers, and estimating wall orientation based on these clusters—could assist bats in pathfinding, even under the assumption of full confusion. At bat densities of 1 to 40 bats/3m² with masking, the multi-call clustering completely restored the collision rate with walls from 0.85 back to 0.2 collisions per second, and significantly improved the exit probability, raising it to 58%, although it did not entirely eliminate the impact of confusion. Our assumption of total confusion between echoes from a bat's own calls and those from conspecifics, as well as our relatively simple clustering model, likely underestimates the true capabilities of real bats when flying in complex environments.

Navigation in bats involves processing complex sensory inputs and applying effective decision-making, often requiring an ability to switch strategies^{77–83}. Bats possess a highly accurate spatial memory^{71,79,83–85}, which is essential for tasks like long-distance migration⁵¹, homing⁸⁶, and maneuvering in cluttered environments⁸⁴. Additionally, they utilize acoustic landmarks to orient in total darkness⁵², occasionally rely on vision^{80,81}, particularly at the cave edge where light is available, can passively detect echolocating peers, and perhaps eavesdrop on conspecifics' echoes²³. In this study we focused on whether echolocation alone is sufficient for one of the most difficult orientation tasks that bats perform – exiting a roost at high densities without prior knowledge of the roost's shape, aside from the initial flight direction. Thus, our echolocation-only model, which was based on a five-call integration window during most simulations, probably underestimates real bats' actual performance which also benefits from additional sensory input and can employ additional navigation strategies by sharing information between each other to coordinate and optimize the routes, such as manifested by swarming intelligence^{33,87,88}.

Our model highlights the importance of considering sensory interference in animal behavior research and illuminates the impressive capabilities of echolocating bats. Additionally, the model showcases the value of simulations and establishes a framework for future studies on collective movement and swarming animals, and on robotics in complex environments.

Methods

The simulated bats rely solely on echolocation to detect and locate obstacles and other bats by analyzing the sound waves they receive. They emit directional echolocation calls and receive the echoes reflected by roost walls and conspecifics, as well as the calls of conspecifics and the echoes returning from their calls. The bats adjust their flight trajectory and echolocation behavior based on the estimated location of the detected objects (range and angle), which deteriorates upon acoustic interference. The detection of the received signals is based on the mammalian gammatone filter bank receiver, under the assumption that bats can differentiate between the desired detected obstacles, conspecifics' echoes, and masking signals. We conducted 2D simulations with varying number of bats (from 1 to 100) to analyze the flight trajectories with and

without masking interference by conspecifics. In the trials without masking interference the bats successfully detected walls and conspecifics without any hindrance. While real-world bat navigation occurs in 3D space, the 2D framework represents a worst-case scenario for echolocation-based navigation, as it increases effective bat density and limits maneuverability compared to a full 3D environment. This approach provides a conservative test of jamming and collision avoidance while maintaining computational tractability, allowing for extensive simulation runs to explore different variables systematically. For a detailed description of the MATLAB simulation see Mazar & Yovel 2020⁷.

The simulation arena was designed to mimic a roost with a corridor-like layout, measuring 14.5 meters in length and 2.5 meters in width, featuring a right-angle turn located 5.5 meters before the exit (see **Figure 1A** for a top-down view). All bats started at a random position within a 2×1.5 m area at the far end of the cave, each initiating flight within a 0.1-second window in a random direction between -30° and $+30^\circ$ relative to the exit (see **Figure 1**). They employ a simple navigation algorithm that dynamically adjusts flight direction based on the detected obstacles or conspecifics (Supplementary Figure 1 and **Figure 1D**). If no obstacles or conspecifics are detected, they continue in a correlated random walk with a maximal turning rate of approximately 30 deg/sec. When obstacles are detected, they are first localized with an error (see below and⁷). Then, if an opening (i.e., a gap of at least 0.5 m between obstacles) is detected, the bats fly through it, if not, they follow the walls while maintaining a 0.8 m distance from them. When approaching an obstacle too closely (<1.5 m and at an angle $<60^\circ$), they execute an obstacle avoidance maneuver. Close proximity to another bat (<0.4 m) triggers an avoidance maneuver away from the nearest conspecific. To evaluate the choice of these distances (1.5 m from walls and 0.4 m from other bats), we tested the sensitivity of the model to conspecific avoidance distances ranging from 0.2 to 1.6 meters across bat densities of 2 to 40 bats/3m². We observed only a modest effect on exit probability at the highest density, where exit probability increased slightly from 82% to 88% ($p = 0.024$, $t = 2.25$, $DF = 958$). No significant changes were observed in exit time, collision rate, or jamming probability across other densities or conditions (GLM, with the number of bats and avoidance distance set as fixed explanatory variables, and the outcome variable being one of: exit probability, time-to-exit, collision rate, or jamming probability). These findings confirm that the modeled behavior is largely insensitive to this parameter range.

If the bat collides with a wall, it immediately turns so that its new flight direction is at a 90° angle to the wall. Collisions between conspecifics, which are common in nature and generally not disruptive in low velocities, are not explicitly modeled. Instead, during the collision event the bat keeps decreasing its velocity and changing its flight direction until the distance between bats is above the threshold (0.4 m). We assume that the primary cost of such interactions arises from the effort required to avoid collisions resulting in forced changes in flight's direction and speed, rather than from the collision itself. Each decision relies on a multi-call integration window that records the estimated locations of detected reflectors from recent echolocation calls (see Supplementary Figure 2A). By default, this window includes the last five calls, and we systematically tested the effect of using between 1 and 10 calls. This algorithm functions without any prior knowledge of the bats' location or the roost's structure. To assess performance, we measured the probability of successfully exiting the roost within a 15-second window. The time-based exit limit was chosen because it is approximately twice the average exit time for 40 bats under acoustic interference in our model, allowing bats sufficient time to correct their trajectory and circle back if they missed the exit on the first attempt. This threshold keeps simulation times reasonable while still capturing the key aspects of exit dynamics.

Echolocation behavior and flight speed follow the phases widely reported in insectivorous bats, categorized as “search,” “approach,” and “buzz”^{89–94} with specific echolocation parameters for *Pipistrellus kuhlii* (Kuhl's pipistrelle)⁵⁹ and *Rhinopoma microphyllum* (greater mouse-tailed bat)²⁵. The transition distances between these phases were identical for both species (see **Table 2**) and are based on empirical studies documenting hunting and obstacle avoidance

behavior^{58,89,93–96}. In nature, call parameters (Inter Pulse Interval (IPI), call duration, and start and stop frequencies) are primarily shaped by the target distance and echo strength. Accordingly, there is little difference in echolocation between prey capture and obstacles-related maneuvers, aside from intensity adjustments based on target strength^{76,96–98}. In our study, due to the dense cave environment, the bats are found to operate in the approach phase nearly all of the time, which is consistent with natural cave emergence behavior, where they are navigating through a cluttered environment rather than engaging in open-space search. Our model was designed to remain as simple as possible while relying on conservative assumptions that may underestimate bat performance. If, in reality, bats fine-tune their echolocation calls even earlier or more precisely during navigation than assumed, our model would still conservatively reflect their actual capabilities.

The simulated echolocation call consists of the dominant harmony of the bat's FM Chirp (1st harmony of the PK and 2nd harmony of the RM). The echolocation signals used in our simulation were modeled as logarithmic FM chirps, implemented using the MATLAB built-in function (e.g., *chirp(t, f0, t1, f1, 'logarithmic')*). This approach aligns with the known nonlinear frequency modulation characteristics of *Pipistrellus kuhlii* (PK) and *Rhinopoma microphyllum* (RM). **Table 2** provides the specific echolocation parameters used in the model, based on Kalko 1995⁵⁸, and Goldshtein 2025²⁵. During the search phase, the bats fly at a nominal velocity of 6 m/sec, reducing it by half during the approach phase and continuously adjusting their speed according to the relative direction of the target, using a delayed linear adaptive law^{7,93,99}. The maneuverability of the bats is constrained to a maximum of 4 m/sec², limiting both angular and linear accelerations. Additionally, our model includes random individual variations in terminal frequencies, assuming a normal distribution with a standard deviation of 1 kHz across the bats.

The sound intensity of the echoes generated by the bat's own calls and those of its conspecifics are calculated using the sonar equation^{7,100} (pp. 196–198), as shown in Equation 1, geometrical relations are according to Supplementary Figure 5. The received levels of the masking calls are determined by using the Friis transmission equation¹⁰¹, as shown in Equation 2. All signal levels were simulated and reported in dB-SPL, referenced to 0.1 meters from the emitting bat. Bats are modeled acoustically as spherical reflectors with a fixed target strength of –23dB assuming reference distance 1 meter, reflecting sound isotropically. This approximates a sphere with a radius of 0.15 m, corresponding to the approximate wingspan of *Rhinopoma microphyllum* (RM)^{25,102}. While target strength can vary with wing posture and body geometry, we chose a representative value within the reported biological range for simplicity and model consistency. Our own measurement of a 3D-printed RM bat yielded a target strength of –32 dB, and a sensitivity analysis (Supplementary Figure 4) showed that model performance was only mildly affected across a wide range of target strengths (see Supplementary Figure 4). This supports the robustness of our approach to different sized bats. Walls are modeled as composites of individual reflectors placed 20 cm apart; each treated as a sphere with a 20 cm radius and a target strength of –22.5dB. For simplicity, in our model, the head is aligned with the body, therefore the direction of the echolocation beam is the same as the direction of the flight. The directivity of the calls and the received echoes is defined by the piston model^{7,92} with radii of 3 mm for the mouth-gap and 7 mm for the ear. The directivity is not directly influenced by velocity but follows behavioral dependent frequency changes. As the bat transitions from search to approach to buzz phases, it emits higher-frequency calls, leading to increased directivity. This shift coincides with a natural reduction in speed during the approach phase. Echo delays are calculated as the two-way travel time of the signals from the emitter to the target.

$$\text{Equation 1: } P_r = P_t \cdot \frac{G_t(\phi_{\text{target}}, f) G_r(\phi_{\text{target}}, f) \lambda^2}{(4\pi)^3 D^4} \cdot 10^{-2\alpha_{\text{att}}(f)/10 \cdot (D-0.1)} \cdot \sigma(f)$$

$$\text{Equation 2: } P_{\text{mask}} = P_t G_t(\phi_{t_x, r_x}, f) G_r(\phi_{r_x, t_x}, f) \cdot \left(\frac{\lambda}{4\pi D_{\text{txrx}}}\right)^2 10^{-\alpha_{\text{att}}(D-0.1)}$$

$$\text{Equation 3: } P_{\text{echoesFromMasking}} = P_t \cdot \frac{G_t(\phi_{t_x}, f) G_r(\phi_{r_x}, f) \lambda^2}{(4\pi)^3 D_{t_x}^2 D_{r_x}^2} 10^{-\alpha_{\text{att}}(D_{t_x} + D_{r_x} - 0.2)} \cdot \sigma(f)$$

where,

P_r : level of the received signal [SPL]

P_t : level of the transmitted call [SPL]

P_{mask} : level of the masking signal as received by the bat [SPL]

$P_{echoesFromMasking}$: level of the echoes reflected by conspecifics and received by the bat [SPL]

$G_t(\Phi, f)$: gain of the transmitter (mouth of the bat, piston model), as a function of azimuth and frequency (f) [numeric]

$G_r(\Phi, f)$: gain of the receiver (ears of the bat, piston model) [numeric]

Φ_{target} : the angle between the bat and the reflected object [rad]

D : distance between the bat and the target [m]

Φ_{txrx}, D_{txrx} : the angle [rad], and the distance [m] between the transmitting conspecific and the receiving focal bat (from the transmitter's perspective)

Φ_{rxtx}, D_{rxtx} : the angle [rad], and the distance [m] between the receiving bat and the transmitting bat (from the receiver's perspective)

Φ_{tx} : the angle [rad], between the masking bat and target (from the transmitter's perspective)

$\alpha_{att}(f)$: atmospheric absorption coefficient for sound [dB/m]

$\sigma(f)$: SONAR cross-section of the target [m^2]

λ : The wavelength of the signal [m]

To maintain model simplicity, we did not incorporate Doppler effects in the echolocation model. While Doppler shifts can affect frequency perception, their impact on jamming and navigation performance is minimal in this context¹⁰³. Moreover, the inter-individual random signals frequencies were larger than the expected Dopplers. In addition, the model does not assign echoes to earlier calls if their delays exceed the bat's own Inter-Pulse Interval (IPI), and thus does not simulate pulse-echo ambiguity.

To model the detection process in the bat's cochlea, we employed a monaural filter bank receiver^{47,104,105} consisting of 80 channels, each with three components: (i) a gammatone filter of order 8, acting as a bandpass filter with center frequencies logarithmically scaled between 10kHz and 80kHz⁷; (ii) a half-wave rectifier; and (iii) a lowpass filter (Butterworth, $f_c=8$ kHz, order=6). Object detection and distance estimation are conducted using Saillant's method^{7,47,106}, based on the sum of detections in the active channels, see **Figure 1C**, D. Initially, a de-chirping process calculates the reference frequency-delay by detecting the peak in the response of each channel to the emitted call in a noise-free environment. Subsequently, the received signal, containing both desired echoes and masking sounds, passes through the filter bank. In each channel, all peaks above a threshold level are detected and time-shifted by the de-chirp reference. The detection threshold in each channel was set to the higher of two values: either 7 dB above the noise floor (0 dB-SPL) or the maximum received signal level minus 70 dB, thereby enforcing a dynamic range of 70dB. Peaks from all channels are aggregated in 5 μ s

windows and convolved with a Gaussian kernel with $\sigma=5\ \mu\text{s}$. Output peaks that exceed the threshold level, set at 10% of the number of active channels, and fall within a time window of $100\ \mu\text{s}$ around the expected delay, are considered successful detections.

To evaluate the impact of acoustic interference, we conducted the detection procedure twice. The first, termed “interference-free detection”, comprised only the desired echoes, with white Gaussian noise at a level of 0 dB-SPL and without masking signals. The second, termed “full detection” comprised the desired echoes, Gaussian noise, and the masking signals. Detected echoes in the full detection were defined by the strongest peak within a four-millisecond window (three milliseconds before and one millisecond after, accounting for forward and backward masking [24](#), [107](#)–[109](#)) detected above the threshold within $100\ \mu\text{s}$ of the interference-free detections. If the detected peak in the full detection condition was delayed by more than $100\ \mu\text{s}$ compared to the interference-free case, it was defined as a miss-detection. Peaks with smaller timing shifts were considered **detection with timing errors**. **Jammed echoes** were defined as echoes that were detected under the interference-free condition but not detected under the full detection condition. The **jamming probability** was calculated as the ratio of jammed echoes in the full detection condition to the detected echoes in the interference-free condition.

After detection, the bat estimates the range and the Direction of Arrival (DOA) of the reflecting objects. The range is determined by the delay of the detected echo, including any errors derived from the filter-bank process in the “full detection” process (i.e., including all masking signals) [7](#), [110](#), [113](#). The direction is not explicitly estimated through binaural processing. Instead, based on previous studies [107](#), [110](#), we assumed that bats can estimate the direction of arrival with an angular error that depends on the Signal-to-Noise Ratio (SNR) and the angle. The inputs to this process include the peak level of the desired echo, the noise level, and the level of acoustic interference. The output is the estimated direction of arrival with a random error applied based on the SNR. At an angle of 0° and an SNR of 10 dB, the standard deviation of the error is 1.5° [111](#) and [7](#) (Equation 1 [0](#)).

To evaluate the impact of the assumption that bats can distinguish between echoes caused by their own calls and those caused by other bats (i.e., conspecifics’ reflectors), we tested an alternative model in which the simulated bats treat all echoes reflected from walls as if they have originated from their own calls. The distance to reflectors of conspecifics’ calls is estimated based on the time difference between the echo and the bat’s last call. The direction of arrival is estimated by the angle between the bat and the physical reflector, with an added random error (the same process used for their own echoes).

In real bats, spatial processing in the brain involves integrating auditory and spatial information over time to construct a coherent map of the environment [5](#), [57](#). This neural computation is crucial for navigation and prey detection in complex environments. To examine whether spatial integration mitigates the confusion problem, we added a ‘multi-call clustering’ module that was based on the sensory information obtained within a one-second memory window. The clustering comprised the following steps: (i) clustering all detections in memory into groups with a maximum internal distance of 10 cm; (ii) reconstructing the estimated walls positions and directions based on the average of clusters that include at least two detections (rather than relying on single reflections); and (iii) identifying openings between reconstructed wall edges ranging from 0.5 to 2.25 meters in width, see Supplementary Figure 1 and Supplementary Figure 2B. The model assumes that bats store echo locations in an allocentric x-y coordinate system, transforming detections from a local to a global spatial framework. Collision avoidance is based not only on the integrated spatial representation but also on immediate echoes from the last call (prior to clustering), including potential uncorrected false detections and localization errors, which are independently processed for real-time evasive maneuvers.

Statistical analysis

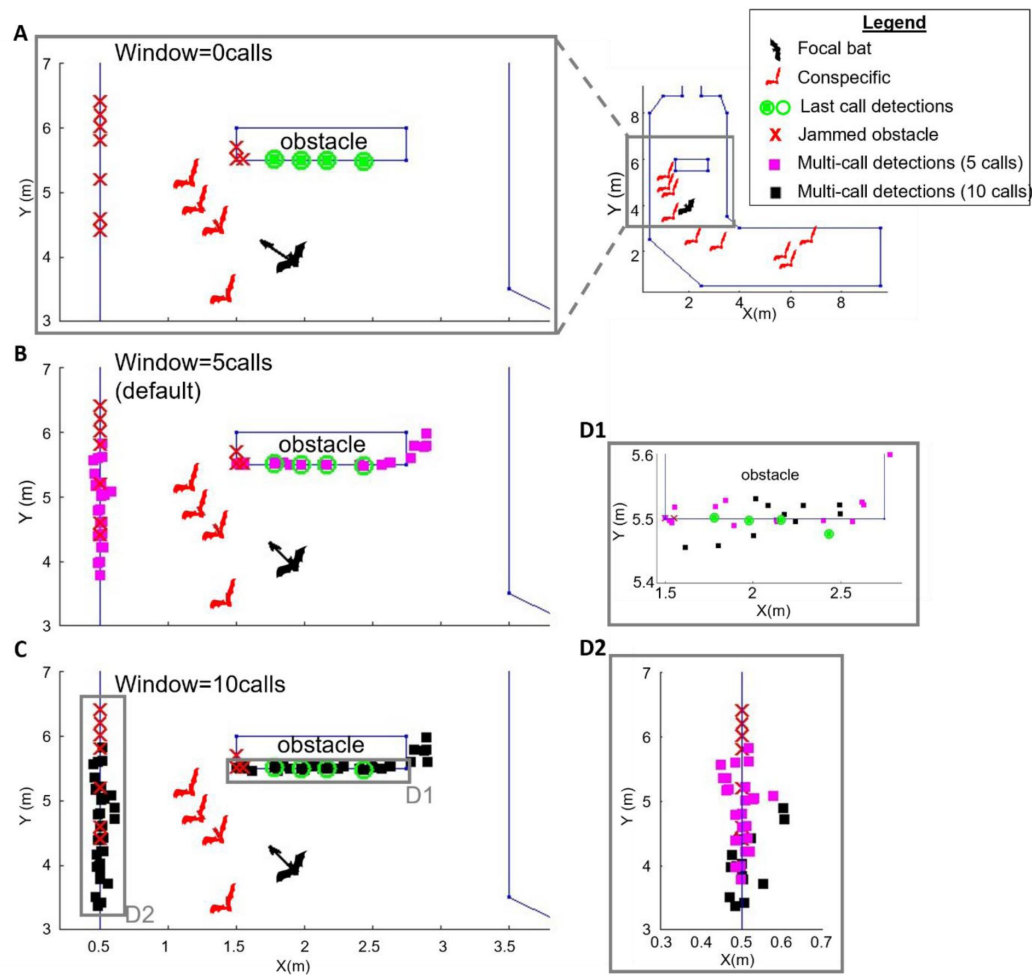
Statistical analysis and the roost-exit model were conducted using MATLAB© 2023a.

Tests were performed with a significance level of 0.05. For each simulated scenario, we examined the effect of the various parameters on exit probability, time-to-exit, collision rate, and the jamming probability, using Generalized Linear Models (GLMs). The GLM tests were executed with MATLAB built-in function ‘**fitglm()**’. Probability variables (such as exit and jamming probabilities) were treated as binomially distributed; rate variables (such as collision rate) were treated as Poisson distributed, and all other variables were considered normally distributed. Unless otherwise stated, all explaining variables were set as fixed factors. All statistical analyses, including the statistical test and the corresponding sample sizes, are described throughout the text and summarized in **Table 1** [↗](#). Standard errors are calculated across all individuals within each scenario, without distinguishing between different simulation trials.

Data availability

All data and codes generated during this study are included in the manuscript and supporting files. Source code files have been uploaded with a Graphical User Interface and a readme file for explanation. Data are available at: <https://github.com/omermazar/Colony-Exit-Bat-Simulation/tree/main> [↗](#)

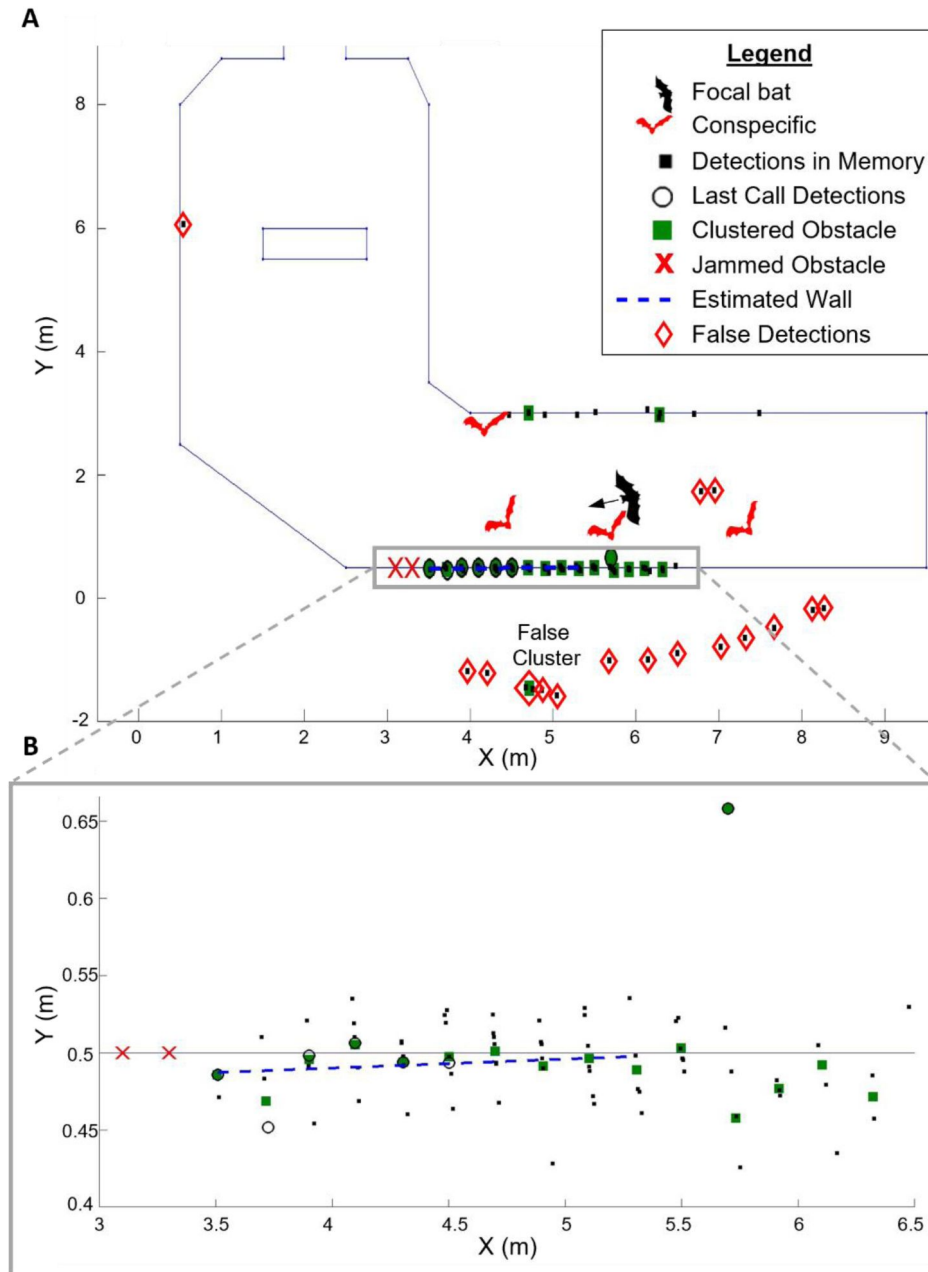
Supplementary material



Supplementary Figure 2A

Multi-Call Integration

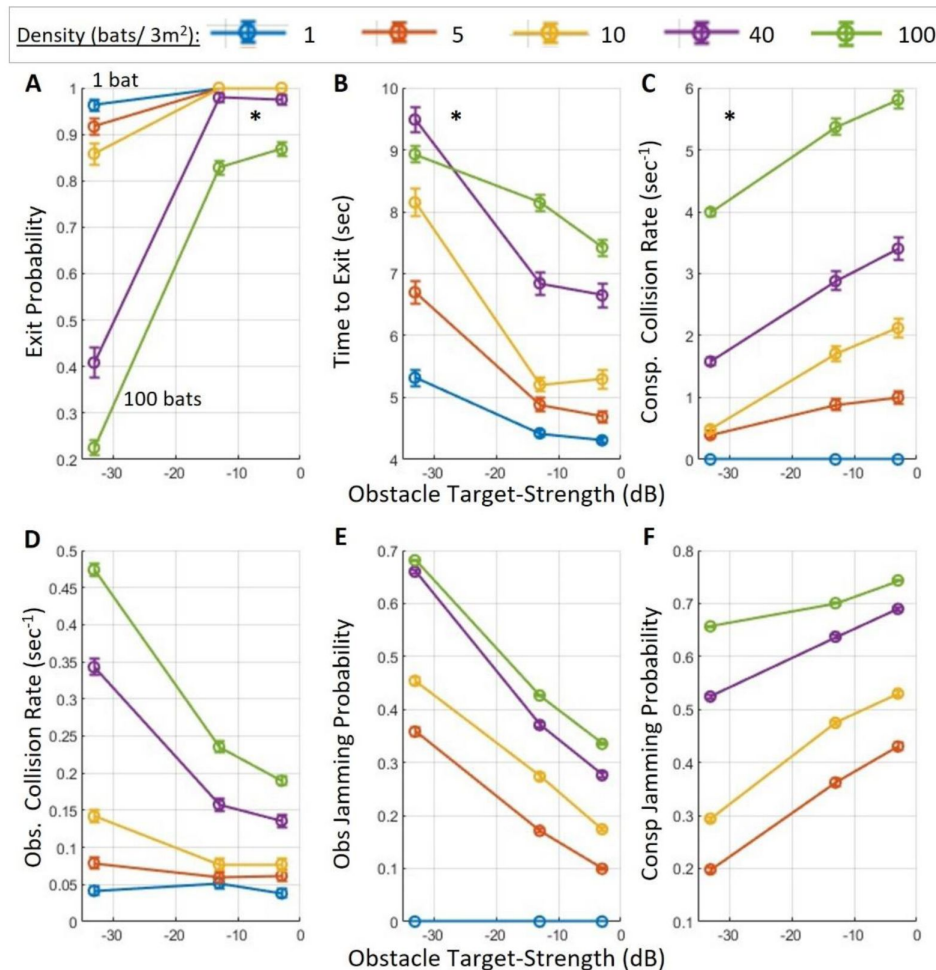
This figure demonstrates the effect of multi-call integration under non-confusing conditions. The upper-right panel shows the position of the focal bat (black) and nine conspecifics (red) within the roost corridor, with a zoomed-in view of the gray rectangle provided in Panels A–C. **(A)** When the integration window is set to zero calls (no memory), the bat relies solely on the latest call. Green circles and squares represent detected reflectors, while red Xs indicate missed (jammed) detections. Notably, the left wall of the corridor remains undetected due to jamming. **(B, C)** Increasing the integration window to five calls (magenta squares) and ten calls (black squares) accumulates detections from prior calls, improving coverage of the environment. In this basic integration model, each detection is treated independently, without clustering. **(D1, D2)** Magnified views of the gray regions indicated in Panel C, comparing detections across 0, 5, and 10-call windows (green, magenta, and black, respectively), illustrating how extended memory improves detection robustness. Note that the X-Y aspect ratios in D1 and D2 differ from the main panels to enhance visibility of spatial distributions.



Supplementary Figure 2B

Multi-Call Clustering Example

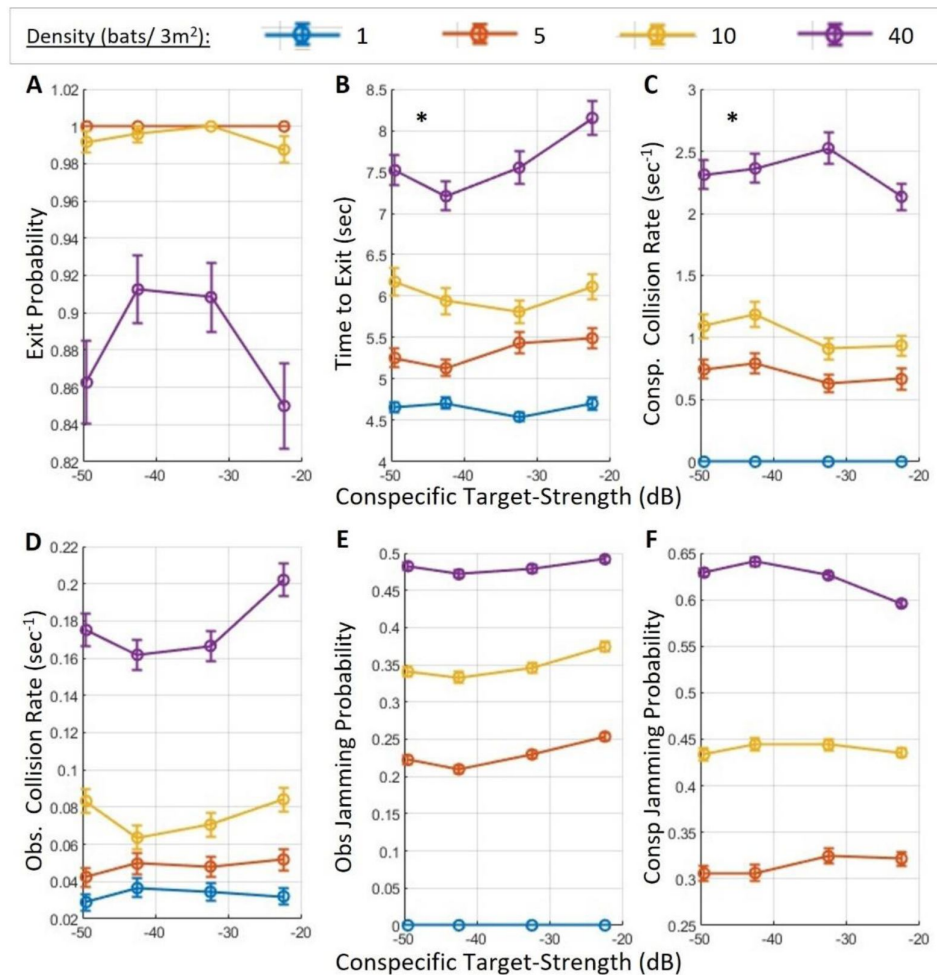
This figure illustrates the multi-call clustering algorithm under full-confusion conditions. (A) A focal bat (black) and four conspecifics (red) are shown in the lower corridor. (B) A zoom-in of the gray rectangle in (A). Black ovals represent detections from the last call; red X's indicate jammed echoes; black squares represent all detections stored across the integration window (before clustering), each subject to localization error. When not applying multi-call clustering – the bat would rely on all of these dots as reflectors. Under full confusion, the bat cannot distinguish self-echoes from conspecific echoes, leading to false detections (red diamonds). Detections are clustered when a reflector is detected twice or more within a 10 cm radius (green squares). The clustered reflectors are used to estimate wall directions (blue dashed line) and detect possible gaps (not shown). As a result of the multi-call clustering algorithm, most false detections are removed as outliers, except for one erroneous cluster (Panel A). Collision avoidance maneuvers are based on both the clustered obstacles and the raw detections from the latest call (empty black ovals).



Supplementary Figure 3

Sensitivity of exit performance to obstacle target strength

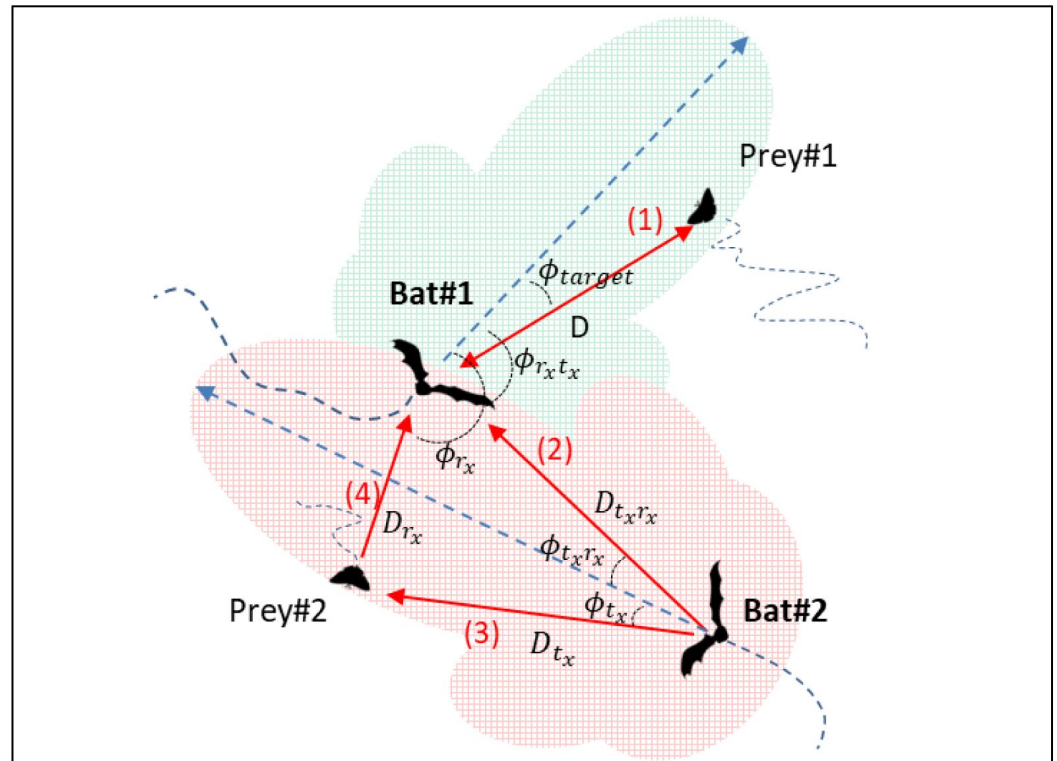
This figure shows how changes in the acoustic target strength of the cave walls affect navigation performance across five bat densities (1, 5, 10, 40, and 100 bats/3 m²). Target strength values ranged from -33 dB to -3 dB, corresponding to spherical reflectors with approximate radii from 0.05 m to 1.5 m. Overall, increasing obstacle target strength significantly influenced exit performance, primarily by reducing the probability of obstacle jamming and thereby improving detection. **(A) Exit Probability** increased with obstacle target strength across all densities, with a maximal increase of 64% for a density of 100 bats ($p < 10^{-10}$, $t = 28.5$, $DF = 8157$, GLM). **(B) Time to Exit** decreased significantly with increasing obstacle target strength, with a maximal reduction of approximately 2.8 seconds at a density of 10 bats ($p < 10^{-10}$, $t = -22.2$, $DF = 6920$, GLM). **(C) Conspecific Collision Rate** increased slightly with stronger obstacle echoes ($p < 10^{-10}$, $t = 27.6$, $DF = 8157$, GLM). **(D) Observed Collision Rate** decreased significantly with increasing target strength ($p < 10^{-10}$, $t = -10.7$, $DF = 8157$, GLM), reflecting better detection of walls and structures. **(E) Observed Jamming Probability** decreased consistently ($p < 10^{-10}$, $t = -19.8$, $DF = 8157$, GLM). **(F) Conspecific Jamming Probability** increased with obstacle target strength ($p < 10^{-10}$, $t = 27.6$, $DF = 8157$, GLM). These results suggest that stronger wall echoes improve environmental awareness at the cost of slightly increased masking of conspecific echoes. Despite this, the overall performance— particularly exit probability and reduced obstacle collisions—improves significantly. In all panels, circles represent means and bars represent standard errors. The error bars are present but very small due to the large number of simulation repetitions, and thus may not be visually noticeable at the plotted scale. See [Table 1](#) for the number of simulated bats.



Supplementary Figure 4

Sensitivity of exit performance to conspecific's target strength

This figure shows how changes in the acoustic target strength of conspecifics affect navigation performance across four bat densities (1, 5, 10, and 40 bats/3 m²). Overall, our results indicate that target strength has a relatively minor impact on performance, likely because it affects both desired echo signals and masking signals equally. Interestingly, this analysis also suggests that our model is more sensitive to the bat's response to nearby conspecifics than to the physical collision impact itself. **(A)** Exit probability was not significantly affected by conspecific target strength ($p=0.28$, $t=-1.09$, $DF=5757$, GLM, see details in [Table 1](#)). Note that the performance curves for densities of 1 and 5 bats overlap almost completely. **(B)** Time-to-exit increased with target strength at high density, with a maximal effect size of ~1 second at 40 bats ($p = 0.003$, $t = 3.02$, $DF = 5578$). **(C, D)** Collision rates with conspecifics decreased significantly with stronger target strength ($p = 0.0002$, $t = -3.7$, $DF = 5757$), while collisions with obstacles remained statistically unchanged ($p = 0.23$, $t = 1.18$, $DF = 5757$). **(E, F)** Jamming probability was not significantly affected for either conspecific or obstacle echoes ($p = 0.6$, $t = -0.51$, $DF = 4762$; $p = 0.19$, $t = 1.31$, $DF = 5757$, respectively). This aligns with the notion that both useful and interfering signals scale similarly with target strength. Importantly, the probability of detecting a conspecific located within 1 meter increased substantially with higher target strength, improving from 25% to 43% at 40 bats ($p < 10^{-10}$, $t = 6.45$, $DF = 4162$). In all panels, circles represent means and bars represent standard errors. The error bars are present but very small due to the large number of simulation repetitions, and thus may not be visually noticeable at the plotted scale. See [Table 1](#) for the number of simulated bats.



Supplementary Figure 5

Angles and distances for two bats and two reflecting objects.

Bat1 receives a reflected echo from Prey1 or a stationary obstacle located at a distance of D from it, with an angle ϕ_{target} relative to its flight direction (red arrow 1). Prey1 is also within the detection range of Bat1, depicted by the green shaded piston area. Bat1 also receives masking sounds from Bat2. The echolocation signals emitted by Bat2 arrive at the ear of Bat1 at an angle ϕ_{txrx} relative to its flight direction and from a distance of D_{txrx} (red arrow 2). Additionally, the echolocation signals of Bat2 are reflected by Prey2, before being received by Bat 1. These reflected signals act as masking signals at a relative angle of angle ϕ_{rx} , and from a distance of D_{rx} from Bat1.

Additional files

Supplementary Movie 1 [🔗](#)

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Reviewer #1 (Public review):

Summary:

Mazar & Yovel 2025 dissect the inverse problem of how echolocators in groups manage to navigate their surroundings despite intense jamming using computational simulations.

The authors show that despite the 'noisy' sensory environments that echolocating groups present, agents can still access some amount of echo-related information and use it to navigate their local environment. It is known that echolocating bats have strong small and large-scale spatial memory that plays an important role for individuals. The results from this paper also point to the potential importance of an even lower-level, short-term role of memory in the form of echo 'integration' across multiple calls, despite the unpredictability of echo detection in groups. The paper generates a useful basis to think about the mechanisms in echolocating groups for experimental investigations too.

Strengths:

- * The paper builds on biologically well-motivated and parametrised 2D acoustics and sensory simulation setup to investigate the various key parameters of interest
- * The 'null-model' of echolocators not being able to tell apart objects & conspecifics while echolocating still shows agents successfully emerge from groups - even though the probability of emergence drops severely in comparison to cognitively more 'capable' agents. This is nonetheless an important result showing the direction-of-arrival of a sound itself is the 'minimum' set of ingredients needed for echolocators navigating their environment.
- * The results generate an important basis in unraveling how agents may navigate in sensorially noisy environments with a lot of irrelevant and very few relevant cues.
- * The 2D simulation framework is simple and computationally tractable enough to perform multiple runs to investigate many variables - while also remaining true to the aim of the investigation.

Weaknesses:

* Authors have not yet provided convincing justification for the use of different echolocation phases during emergence and in cave behaviour. In the previous modelling paper cited for the details - here the bat-agents are performing a foraging task, and so the switch in echolocation phases is understandable. While flying with conspecifics, the lab's previous paper has shown what they call a 'clutter response' - but this is not necessarily the same as going into a 'buzz'-type call behaviour. As pointed out by another reviewer - the results of the simulations may hinge on the fact that bats are showing this echolocation phase-switching, and thus improving their echo-detection. This is not necessarily a major flaw - but something for readers to consider in light of the sparse experimental evidence at hand currently.

* The decision to model direction-of-arrival with such high angular resolution (1-2 degrees) is not entirely justifiable - and the authors may wish to do simulation runs with lower angular resolution. Past experimental paradigms haven't really separated out target-strength as a confounding factor for angular resolution (e.g. see the cited Simmons et al. 1983 paper). Moreover, to this reviewer's reading of the cited paper - it is not entirely clear how this experiment provides source-data to support the DoA-SNR parametrisation in this manuscript. The cited paper has two array-configurations, both of which are measured to have similar received levels upon ensonification. A relationship between angular resolution and signal-to-noise ratio is understandable perhaps - and one can formulate such a relationship, but here the reviewer asks that the origin/justification be made clear. On an independent line, also see the recent contrasting results of Geberl, Kugler, Wiegrebe 2019 (Curr. Biol.) - who suggest even poorer angular resolution in echolocation.

<https://doi.org/10.7554/eLife.105571.2.sa2>

Reviewer #2 (Public review):

This manuscript describes a detailed model for bats flying together through a fixed geometry. The model considers elements which are faithful to both bat biosonar production and reception and the acoustics governing how sound moves in air and interacts with obstacles. The model also incorporates behavioral patterns observed in bats, like one-dimensional feature following and temporal integration of cognitive maps. From a simulation study of the model and comparison of the results with the literature, the authors gain insight into how often bats may experience destructive interference of their acoustic signals and those of their peers, and how much such interference may actually negatively effect the groups' ability to navigate effectively. The authors use generalized linear models to test the significance of the effects they observe.

The work relies on a thoughtful and detailed model which faithfully incorporates salient features, such as acoustic elements like the filter for a biological receiver and temporal aggregation as a kind of memory in the system. At the same time, the authors abstract features that are complicating without being expected to give additional insights, as can be seen in the choice of a two-dimensional rather than three-dimensional system. I thought that the level of abstraction in the model was perfect, enough to demonstrate their results without needless details. The results are compelling and interesting, and the authors do a great job discussing them in the context of the biological literature.

With respect to the first version of the manuscript, the authors have remedied all my outstanding questions or concerns in the current version. The new supplementary figure 5 is especially helpful in understanding the geometry.

<https://doi.org/10.7554/eLife.105571.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

We thank the reviewer for his valuable input and careful assessment, which have significantly improved the clarity and rigor of our manuscript.

Summary:

Mazer & Yovel 2025 dissect the inverse problem of how echolocators in groups manage to navigate their surroundings despite intense jamming using computational simulations.

The authors show that despite the 'noisy' sensory environments that echolocating groups present, agents can still access some amount of echo-related information and use it to navigate their local environment. It is known that echolocating bats have strong small and large-scale spatial memory that plays an important role for individuals. The results from this paper also point to the potential importance of an even lower-level, short-term role of memory in the form of echo 'integration' across multiple calls, despite the unpredictability of echo detection in groups. The paper generates a useful basis to think about the mechanisms in echolocating groups for experimental investigations too.

Strengths:

- (1) The paper builds on biologically well-motivated and parametrised 2D acoustics and sensory simulation setup to investigate the various key parameters of interest*
- (2) The 'null-model' of echolocators not being able to tell apart objects & conspecifics while echolocating still shows agents successfully emerge from groups - even though the probability of emergence drops severely in comparison to cognitively more 'capable' agents. This is nonetheless an important result showing the direction-of-arrival of a sound itself is the 'minimum' set of ingredients needed for echolocators navigating their environment.*
- (3) The results generate an important basis in unraveling how agents may navigate in sensorially noisy environments with a lot of irrelevant and very few relevant cues.*
- (4) The 2D simulation framework is simple and computationally tractable enough to perform multiple runs to investigate many variables - while also remaining true to the aim of the investigation.*

Weaknesses:

There are a few places in the paper that can be misunderstood or don't provide complete details. Here is a selection:

- (1) Line 61: '... studies have focused on movement algorithms while overlooking the sensory challenges involved' : This statement does not match the recent state of the literature. While the previous models may have had the assumption that all neighbours can be detected, there are models that specifically study the role of limited interaction arising from a potential inability to track all neighbours due to occlusion, and the effect of responding to only one/few neighbours at a time e.g. Bode et al. 2011 R. Soc. Interface, Rosenthal et al. 2015 PNAS, Jhawar et al. 2020 Nature Physics.*

We appreciate the reviewer's comment and the relevant references. We have revised the manuscript accordingly to clarify the distinction between studies that incorporate limited interactions and those that explicitly analyze sensory constraints and interference. We have refined our statement to acknowledge these contributions while maintaining our focus on sensory challenges beyond limited neighbor detection, such as signal degradation, occlusion effects, and multimodal sensory integration (see lines 58-64):

(2) The word 'interference' is used loosely places (Line 89: '...took all interference signals...', Line 319: 'spatial interference') - this is confusing as it is not clear whether the authors refer to interference in the physics/acoustics sense, or broadly speaking as a synonym for reflections and/or jamming.

To improve clarity, we have revised the manuscript to distinguish between different types of interference:

- Acoustic interference (jamming): Overlapping calls that completely obscure echo detection, preventing bats from perceiving necessary environmental cues.
- Acoustic interference (masking): Partial reduction in signal clarity due to competing calls.
- Spatial interference: Physical obstruction by conspecifics affecting movement and navigation.

We have updated the manuscript to use these terms consistently and explicitly define them in relevant sections (see lines 84-85, 119-120). This distinction ensures that the reader can differentiate between interference as an acoustic phenomenon and its broader implications in navigation.

(3) The paper discusses original results without reference to how they were obtained or what was done. The lack of detail here must be considered while interpreting the Discussion e.g. Line 302 ('our model suggests...increasing the call-rate..' - no clear mention of how/where call-rate was varied) & Line 323 '..no benefit beyond a certain level..' - also no clear mention of how/where call-level was manipulated in the simulations.

All tested parameters, including call rate dynamics and call intensity variations, are detailed in the Methods section and Tables 1 and 2. Specifically:

- Call Rate Variation: The Inter-Pulse Interval (IPI) was modeled based on documented echolocation behavior, decreasing from 100 msec during the search phase to 35 msec (~28 calls per second) at the end of the approach phase, and to 5 msec (200 calls per second) during the final buzz (see Table 2). This natural variation in call rate was not manually manipulated in the model but emerged from the simulated bat behavior.
- Call Intensity Variation: The tested call intensity levels (100, 110, 120, 130 dB SPL) are presented in Table 1 under the "Call Level" parameter. The effect of increasing call intensity was analyzed in relation to exit probability, jamming probability, and collision rate. This is now explicitly referenced in the Discussion. We have revised the manuscript to explicitly reference these aspects in the Results and Discussion sections – see lines 346-349, 372-375.

Reviewer #2 (Public review):

We are grateful for the reviewer's insightful feedback, which has helped us clarify key aspects of our research and strengthen our conclusions.

This manuscript describes a detailed model of bats flying together through a fixed geometry. The model considers elements that are faithful to both bat biosonar production and reception and the acoustics governing how sound moves in the air and interacts with obstacles. The model also incorporates behavioral patterns observed in bats, like one-dimensional feature following and temporal integration of cognitive maps. From a simulation study of the model and comparison of the results with the literature, the authors gain insight into how often bats may experience destructive interference of their acoustic signals and those of their peers, and how much such interference may actually negatively affect the groups' ability to navigate effectively. The authors use generalized linear models to test the significance of the effects they observe.

In terms of its strengths, the work relies on a thoughtful and detailed model that faithfully incorporates salient features, such as acoustic elements like the filter for a biological receiver and temporal aggregation as a kind of memory in the system. At the same time, the authors' abstract features are complicating without being expected to give additional insights, as can be seen in the choice of a twodimensional rather than three-dimensional system. I thought that the level of abstraction in the model was perfect, enough to demonstrate their results without needless details. The results are compelling and interesting, and the authors do a great job discussing them in the context of the biological literature.

The most notable weakness I found in this work was that some aspects of the model were not entirely clear to me.

For example, the directionality of the bat's sonar call in relation to its velocity. Are these the same?

For simplicity, in our model, the head is aligned with the body, therefore the direction of the echolocation beam is the same as the direction of the flight.

Moreover, call directionality (directivity) is not directly influenced by velocity. Instead, directionality is estimated using the piston model, as described in the Methods section. The directionality is based on the emission frequency and is thus primarily linked to the behavioral phases of the bat, with frequency shifts occurring as the bat transitions from search to approach to buzz phases. During the approach phase, the bat emits calls with higher frequencies, resulting in increased directionality. This is supported by the literature (Jakobsen and Surlykke, 2010; Jakobsen, Brinkløv and Surlykke, 2013). This phase is also associated with a natural reduction in flight speed, which is a well-documented behavioral adaptation in echolocating bats (Jakobsen et al., 2024).

To clarify this in the manuscript, we have updated the text to explicitly state that directionality follows phase-dependent frequency changes rather than being a direct function of velocity, see lines 543-545.

If so, what is the difference between ϕ_{target} and ϕ_{tx} in the model equations?

ϕ_{target} represents the angle between the bat and the reflected object (target).

ϕ_{Tx} the angle [rad], between the masking bat and target (from the transmitter's perspective)

ϕ_{TxRx} refers to the angle between the transmitting conspecific and the receiving focal bat, from the transmitter's point of view.

ϕ_{RxTx} represents the angle between the receiving bat and the transmitting bat, from the receiver's point of view.

These definitions have been explicitly stated in the revised manuscript to prevent any ambiguity (lines 525-530). Additionally, a Supplementary figure demonstrating the geometrical relations has been added to the manuscript.

What is a bat's response to colliding with a conspecific (rather than a wall)?

In nature, minor collisions between bats are common and typically do not result in significant disruptions to flight (Boerma et al., 2019; Roy et al., 2019; Goldshtein et al., 2025). Given this, our model does not explicitly simulate the physical impact of a collision event. Instead, during the collision event the bat keeps decreasing its velocity and changing its flight direction until the distance between bats is above the threshold (0.4 m). We assume that the primary cost of such interactions arises from the effort required to avoid collisions, rather than from the collision itself. This assumption aligns with observations of bat behavior in dense flight environments, where individuals prioritize collision avoidance rather than modeling post-collision dynamics. See lines 479-484.

From the statistical side, it was not clear if replicate simulations were performed. If they were, which I believe is the right way due to stochasticity in the model, how many replicates were used, and are the standard errors referred to throughout the paper between individuals in the same simulation or between independent simulations, or both?

The number of repetitions for each scenario is detailed in Table 1, but we included it in a more prominent location in the text for clarity. Specifically, we now state (Lines 110-111):

"The number of repetitions for each scenario was as follows: 1 bat: 240; 2 bats: 120; 5 bats: 48; 10 bats: 24; 20 bats: 12; 40 bats: 12; 100 bats: 6."

Regarding the reported standard errors, they are calculated across all individuals within each scenario, without distinguishing between different simulation trials.

We clarified in the revised text (Lines 627-628 in Statistical Analysis)

Overall, I found these weaknesses to be superficial and easily remedied by the authors. The authors presented well-reasoned arguments that were supported by their results, and which were used to demonstrate how call interference impacts the collective's roost exit as measured by several variables. As the authors highlight, I think this work is valuable to individuals interested in bat biology and behavior, as well as to applications in engineered multi-agent systems like robotic swarms.

Reviewer #3 (Public review):

We sincerely appreciate the reviewer's thoughtful comments and the time invested in evaluating our work, which have greatly contributed to refining our study.

We would like to note that in general, our model often simplifies some of the bats' abilities, under the assumption that if the simulated bats manage to perform this difficult task with simpler mechanisms, real better adapted bats will probably perform even better. This thought strategy will be repeated in several of the s below.

Summary:

The authors describe a model to mimic bat echolocation behavior and flight under high-density conditions and conclude that the problem of acoustic jamming is less severe than previously thought, conflating the success of their simulations (as described in the

*manuscript) with hard evidence for what real bats are actually doing. The authors base their model on two species of bats that fly at "high densities" (defined by the authors as colony sizes from tens to tens of thousands of individuals and densities of up to 33.3 bats/m²), *Pipistrellus kuhli* and *Rhinopoma microphyllum*. This work fits into the broader discussion of bat sensorimotor strategies during collective flight, and simulations are important to try to understand bat behavior, especially given a lack of empirical data. However, I have major concerns about the assumptions of the parameters used for the simulation, which significantly impact both the results of the simulation and the conclusions that can be made from the data. These details are elaborated upon below, along with key recommendations the authors should consider to guide the refinement of the model.*

Strengths:

This paper carries out a simulation of bat behavior in dense swarms as a way to explain how jamming does not pose a problem in dense groups. Simulations are important when we lack empirical data. The simulation aims to model two different species with different echolocation signals, which is very important when trying to model echolocation behavior. The analyses are fairly systematic in testing all ranges of parameters used and discussing the differential results.

Weaknesses:

The justification for how the different foraging phase call types were chosen for different object detection distances in the simulation is unclear. Do these distances match those recorded from empirical studies, and if so, are they identical for both species used in the simulation?

The distances at which bats transition between echolocation phases are identical for both species in our model (see Table 2). These distances are based on well-documented empirical studies of bat hunting and obstacle avoidance behavior (Griffin, Webster and Michael, 1958; Simmons and Kick, 1983; Schnitzler et al., 1987; Kalko, 1995; Hiryu et al., 2008; Vanderelst and Peremans, 2018). These references provide extensive evidence that insectivorous bats systematically adjust their echolocation calls in response to object proximity, following the characteristic phases of search, approach, and buzz.

To improve clarity, we have updated the text to explicitly state that the phase transition distances are empirically grounded and apply equally to both modeled species (lines 499-508).

What reasoning do the authors have for a bat using the same call characteristics to detect a cave wall as they would for detecting a small insect?

In echolocating bats, call parameters are primarily shaped by the target distance and echo strength. Accordingly, there is little difference in call structure between prey capture and obstacles-related maneuvers, aside from intensity adjustments based on target strength (Hagino et al., 2007; Hiryu et al., 2008; Surlykke, Ghose and Moss, 2009; Kothari et al., 2014). In our study, due to the dense cave environment, the bats are found to operate in the approach phase most of the time, which is consistent with natural cave emergence, where they are navigating through a cluttered environment rather than engaging in open-space search. For one of the species (*Rhinopoma*), we also have empirical recordings of individuals flying under similar conditions (Goldshtein et al., 2025). Our model was designed to remain as simple as possible while relying on conservative assumptions that may underestimate bat performance. If, in reality, bats fine-tune their echolocation calls even earlier or more precisely during navigation than assumed, our model would still conservatively reflect their actual capabilities. See lines 500-508.

The two species modeled have different calls. In particular, the bandwidth varies by a factor of 10, meaning the species' sonars will have different spatial resolutions. Range resolution is about 10x better for PK compared to RM, but the authors appear to use the same thresholds for "correct detection" for both, which doesn't seem appropriate.

The detection process in our model is based on Saillant's method using a filterbank, as detailed in the paper (Saillant et al., 1993; Neretti et al., 2003; Sanderson et al., 2003). This approach inherently incorporates the advantages of a wider bandwidth, meaning that the differences in range resolution between the species are already accounted for within the signal-processing framework. Thus, there is no need to explicitly adjust the model parameters for bandwidth variations, as these effects emerge from the applied method.

Also, the authors did not mention incorporating/correcting for/exploiting Doppler, which leads me to assume they did not model it.

The reviewer is correct. To maintain model simplicity, we did not incorporate the Doppler effect or its impact on echolocation. The exclusion of Doppler effects was based on the assumption that while Doppler shifts can influence frequency perception, their impact on jamming and overall navigation performance is minor within the modelled context.

The maximal Doppler shifts expected for the bats in this scenario are of ~ 1 kHz. These shifts would be applied variably across signals due to the semi-random relative velocities between bats, leading to a mixed effect on frequency changes. This variability would likely result in an overall reduction in jamming rather than exacerbating it, aligning with our previous statement that our model may overestimate the severity of acoustic interference. Such Doppler shifts would result in errors of 2-4 cm in localization (i.e., 200-400 micro-seconds) (Boonman, Parsons and Jones, 2003).

We have now explicitly highlighted this in the revised version (see 548-581).

The success of the simulation may very well be due to variation in the calls of the bats, which ironically enough demonstrates the importance of a jamming avoidance response in dense flight. This explains why the performance of the simulation falls when bats are not able to distinguish their own echoes from other signals. For example, in Figure C2, there are calls that are labeled as conspecific calls and have markedly shorter durations and wider bandwidths than others. These three phases for call types used by the authors may be responsible for some (or most) of the performance of the model since the correlation between different call types is unlikely to exceed the detection threshold. But it turns out this variation in and of itself is what a jamming avoidance response may consist of. So, in essence, the authors are incorporating a jamming avoidance response into their simulation.

We fully agree that the natural variations in call design between the phases contribute significantly to interference reduction (see our discussion in a previous paper in Mazar & Yovel, 2020). However, we emphasize that this cannot be classified as a Jamming Avoidance Response (JAR). In our model, bats respond only to the physical presence of objects and not to the acoustic environment or interference itself. There is no active or adaptive adjustment of call design to minimize jamming beyond the natural phase-dependent variations in call structure. Therefore, while variation in call types does inherently reduce interference, this effect emerges passively from the modeled behavior rather than as an intentional strategy to avoid jamming.

The authors claim that integration over multiple pings (though I was not able to determine the specifics of this integration algorithm) reduces the masking problem.

Indeed, it should: if you have two chances at detection, you've effectively increased your SNR by 3dB.

The reviewer is correct. Indeed, integration over multiple calls improves signal-tonoise ratio (SNR), effectively increasing it by approximately 3 dB per doubling of observations. The specifics of the integration algorithm are detailed in the Methods section, where we describe how sensory information is aggregated across multiple time steps to enhance detection reliability.

They also claim - although it is almost an afterthought - that integration dramatically reduces the degradation caused by false echoes. This also makes sense: from one ping to the next, the bat's own echo delays will correlate extremely well with the bat's flight path. Echo delays due to conspecifics will jump around kind of randomly. However, the main concern is regarding the time interval and number of pings of the integration, especially in the context of the bat's flight speed. The authors say that a 1s integration interval (5-10 pings) dramatically reduces jamming probability and echo confusion. This number of pings isn't very high, and it occurs over a time interval during which the bat has moved 5-10m. This distance is large compared to the 0.4m distance-to-obstacle that triggers an evasive maneuver from the bat, so integration should produce a latency in navigation that significantly hinders the ability to avoid obstacles. Can the authors provide statistics that describe this latency, and discussion about why it doesn't seem to be a problem?

As described in the Methods section, the bat's collision avoidance response does not solely rely on the integration process. Instead, the model incorporates real-time echoes from the last calls, which are used independently of the integration process for immediate obstacle avoidance maneuvers. This ensures that bats can react to nearby obstacles without being hindered by the integration latency. The slower integration on the other hand is used for clustering, outlier removal and estimation wall directions to support the pathfinding process, as illustrated in Supplementary Figure 1.

Additionally, our model assumes that bats store the physical positions of echoes in an allocentric coordinate system (x-y). The integration occurs after transforming these detections from a local relative reference frame to a global spatial representation. This allows for stable environmental mapping while maintaining responsiveness to immediate changes in the bat's surroundings.

See lines 600-616 in the revised version.

The authors are using a 2D simulation, but this very much simplifies the challenge of a 3D navigation task, and there is an explanation as to why this is appropriate. Bat densities and bat behavior are discussed per unit area when realistically it should be per unit volume. In fact, the authors reference studies to justify the densities used in the simulation, but these studies were done in a 3D world. If the authors have justification for why it is realistic to model a 3D world in a 2D simulation, I encourage them to provide references justifying this approach.

We acknowledge that this is a simplification; however, from an echolocation perspective, a 2D framework represents a worst-case scenario in terms of bat densities and maneuverability:

- **Higher Effective Density:** A 2D model forces all bats into a single plane rather than distributing them through a 3D volume, increasing the likelihood of overlap in calls and echoes and making jamming more severe. As described in the text: the average distance to the nearest bat in our simulation is 0.27m (with 100 bats), whereas reported distances in very dense colonies are 0.5m (Fujioka et al., 2021), as observed in *Myotis grisescens* (Sabot and

Hudson, 1995) and *Tadarida brasiliensis* (Theriault et al., no date; Betke et al., 2008; Gillam et al., 2010)

- **Reduced Maneuverability:** In 3D space, bats can use vertical movement to avoid obstacles and conspecifics. A 2D constraint eliminates this degree of freedom, increasing collision risk and limiting escape options.

Thus, our 2D model provides a conservative difficult test case, ensuring that our findings are valid under conditions where jamming and collision risks are maximized. Additionally, the 2D framework is computationally efficient, allowing us to perform multiple simulation runs to explore a broad parameter space and systematically test the impact of different variables.

To address the reviewer's concern, we have clarified this justification in the revised text and will provide supporting references where applicable (see Methods lines 450-455).

The focus on "masking" (which appears to be just in-band noise), especially relative to the problem of misassigned echoes, is concerning. If the bat calls are all the same waveform (downsweep linear FM of some duration, I assume - it's not clear from the text), false echoes would be a major problem. Masking, as the authors define it, just reduces SNR. This reduction is something like $\sqrt{N}$, where N is the number of conspecifics whose echoes are audible to the bat, so this allows the detection threshold to be set lower, increasing the probability that a bat's echo will exceed a detection threshold. False echoes present a very different problem. They do not reduce SNR per se, but rather they cause spurious threshold excursions (N of them!) that the bat cannot help but interpret as obstacle detection. I would argue that in dense groups the mis-assignment problem is much more important than the SNR problem.

There is substantial literature supporting the assumption that bats can recognize their own echoes and distinguish them from conspecific signals (Schnitzler, Bioscience and 2001, no date; Kazial, Burnett and Masters, 2001; Burnett and Masters, 2002; Kazial, Kenny and Burnett, 2008; Chili, Xian and Moss, 2009; Yovel et al., 2009; Beetz and Hechavarría, 2022)). However, we acknowledge that false echoes may present a major challenge in dense groups. To address this, we explicitly tested the impact of the self-echo identification assumption in our study see Results Figure 1: The impact of confusion on performance, and lines 399-404 in the Discussion.

Furthermore, we examined a full confusion scenario, where all reflected echoes from conspecifics were misinterpreted as obstacle reflections (i.e., 100% confusion). Our results show that this significantly degrades navigation performance, supporting the argument that echo misassignment is a critical issue. However, we also explored a simple mitigation strategy based on temporal integration with outlier rejection, which provided some improvement in performance. This suggests that real bats may possess additional mechanisms to enhance self-echo identification and reduce false detections. See lines 411-420 in the manuscript for further discussion.

We actually used logarithmically frequency modulated (FM) chirps, generated using the MATLAB built-in function `chirp(t, f0, t1, f1, 'logarithmic')`. This method aligns with the nonlinear FM characteristics of *Pipistrellus kuhlii* (PK) and *Rhinopoma microphyllum* (RM) and provides a realistic approximation of their echolocation signals. We acknowledge that this was not sufficiently emphasized in the original text, and we have now explicitly highlighted this in the revised version to ensure clarity (see Lines 509-512 in Methods).

The criteria set for flight behavior (lines 393-406) are not justified with any empirical evidence of the flight behavior of wild bats in collective flight. How did the authors determine the avoidance distances? Also, what is the justification for the time limit of 15

seconds to emerge from the opening? Instead of an exit probability, why not instead use a time criterion, similar to "How long does it take X% of bats to exit?" :

While we acknowledge that wild bats may employ more complex behaviors for collision avoidance, we chose to implement a simplified decision-making rule in our model to maintain computational tractability.

The avoidance distances (1.5 m from walls and 0.4 m from other bats) were selected as internal parameters to support stable and realistic flight trajectories while maintaining a reasonable collision rate. These values reflect a trade-off between maneuverability and behavioral coherence under crowding. To address this point, we added a sensitivity analysis to the revised manuscript. Specifically, we tested the effect of varying the conspecific avoidance distance from 0.2 to 1.6 meters at bat densities of 2 to 40 bats/3m². The only statistically significant impact was at the highest density (40 bats/3m²), where exit probability increased slightly from 82% to 88% ($p = 0.024$, $t = 2.25$, $DF = 958$). No significant changes were observed in exit time, collision rate, or jamming probability across other densities or conditions (GLM, see revised Methods). These results suggest that the selected avoidance distances are robust and not a major driver of model performance, see lines 469-47.

The 15-second exit limit was determined as described in the text (Lines 489-491): "A 15-second window was chosen because it is approximately twice the average exit time for 40 bats and allows for a second corrective maneuver if needed." In other words, it allowed each bat to circle the 'cave' twice to exit even in the most crowded environment. This threshold was set to keep simulation time reasonable while allowing sufficient time for most bats to exit successfully.

We acknowledge that the alternative approach suggested by the reviewer—measuring the time taken for a certain percentage of bats to exit—is also valid. However, in our model, some outlier bats fail to exit and continue flying for many minutes, such simulations would lead to excessive simulation times making it difficult to generate repetitions and not teaching us much – they usually resulted from the bat slightly missing the opening (see video S1. Our chosen approach ensures practical runtime constraints while still capturing relevant performance metrics.

What is the empirical justification for the 1-10 calls used for integration?

The "average exit time for 40 bats" is also confusing and not well explained. Was this determined empirically? From the simulation? If the latter, what are the conditions?

Does it include masking, no masking, or which species?

Previous studies have demonstrated that bats integrate acoustic information received sequentially over several echolocation calls (2-15), effectively constructing an auditory scene in complex environments (Ulanovsky and Moss, 2008; Chili, Xian and Moss, 2009; Moss and Surlykke, 2010; Yovel and Ulanovsky, 2017; Salles, Diebold and Moss, 2020). Additionally, bats are known to produce echolocation sound groups when spatiotemporal localization demands are high (Kothari et al., 2014). Studies have documented call sequences ranging from 2 to 15 grouped calls (Moss and Surlykke, 2010), and it has been hypothesized that grouping facilitates echo segregation.

We did not use a single integration window - we tested integration sizes between 1 and 10 calls and presented the results in Figure 3A. This range was chosen based on prior empirical findings and to explore how different levels of temporal aggregation impact navigation performance. Indeed, the results showed that the performance levels between 5-10 calls integration window (Figure 3A)

Regarding the average exit time for 40 bats, this value was determined from our simulations, where it represents the mean time for successful exits under standard conditions with masking. We have revised the text to clarify these details see, lines 489-491.

Reviewer #1 (Recommendations for the authors):

(1) Data Availability:

As it stands now, this reviewer cannot vouch for the uploaded code as it wasn't accessible according to F.A.I.R principles. The link to the code/data points to a private company's file-hosting account that requires logging in or account creation to see its contents, and thus cannot be accessed.

This reviewer urges the authors to consider uploading the code onto an academic data repository from the many on offer (e.g. Dryad, Zenodo, OSF). Some repositories offer an option to share a private link (e.g. Zenodo) to the folder that can then be shared only with reviewers so it is not completely public.

This is a computational paper, and the credibility of the results is based on the code used to generate them.

The code is available at GitHub as required:

<https://github.com/omermazar/Colony-Exit-Bat-Simulation>

(2) Abstract:

Line 22: 'To explore whether..' - replace 'whether' with 'how'?

The sentence was rephrased as suggested by the reviewer.

(2) Main text:

Line 43: '...which may share...' - correct to '...which share...'; as elegantly framed in the authors' previous work - jamming avoidance is unavoidable because all FM bats of a species still share >90% of spectral bandwidth despite a few kHz shift here and there.

The sentence was rephrased as suggested by the reviewer.

Line 49: The authors may wish to additionally cite the work of Fawcett et al. 2015 (J. Comp. Phys A & Biology Open)

Thank you for the suggestion. We have included a citation to the work of Fawcett et al. (2015) in the revised manuscript.

Line 61: This statement does not match the recent state of the literature. While the previous models may have assumed that all neighbours can be detected, there are models that specifically study the role of limited interaction arising from the potential inability to track all neighbours, and the effect of responding to only one/few neighbours at a time e.g. Bode et al. 2011 R. Soc. Interface, Jhawar et al. 2020 Nature Physics.

We have added citations to the important studies suggested by the reviewer, as detailed in the Public Review above.

Line 89: '..took all interference signals into account...' - what is meant by 'interference signals' - are the authors referring to reflections, unclear.

We have revised the sentence and detailed the acoustic signals involved in the process: self-generated echoes, calls from conspecifics, and echoes from cave walls and other bats evoked by those calls, see lines 99-106.

Figure 1A: The colour scheme with overlapping points makes the figure very hard to understand what is happening. The legend has colours from subfigures B-D, adding to the confusion.

What does the yellow colour represent? This is not clear. Also, in general, the color schemes in the simulation trajectories and the legend are not the same, creating some amount of confusion for the reader. It would be good to make the colour schemes consistent and visually separable (e.g. consp. call direct is very similar to consp. echo from consp. call), and perhaps also if possible add a higher resolution simulation visualisation. Maybe it is best to separate out the colour legends for each sub-figure.

The updated figure now includes clearer, more visually separable colors, and consistent color coding across all sub-panels. The yellow trajectory representing the focal bat's flight path is now explicitly labeled, and we adjusted the color mapping of acoustic signals (e.g., conspecific calls vs. echoes) to improve distinction. We also revised the figure caption accordingly and ensured that the legend is aligned with the updated visuals. These modifications aim to enhance interpretability and reduce ambiguity for the reader.

Figure C3: What is 'FB Channel', this is not explained in the legend.

'FB Channel' stands for 'Filter Bank Channel'. This clarification has been added to the caption of Figure 1.

Figure 3: Visually noticing that the colour legend is placed only on sub-figure A is tricky and readers may be left searching for the colour legend. Maybe lay out the legend horizontally on top of the entire figure, so it stands out?

We have adjusted the placement of the color legend in Figure 3 to improve visibility and consistency.

Line 141: '..the probability of exiting..' - how is this probability calculated - not clear.

We have clarified in the revised text that the probability of exiting the cave within 15 seconds is defined as the number of bats that exited the cave within that time divided by the total number of bats in each scenario, see lines 159-160.

Line 142: What are the sample sizes here - i.e. how many simulation replicates were performed?

We have clarified the number of repetitions in each scenario the revised text, as detailed in the Public Review above.

Line 151: 'The jamming probability,...number of jammed echoes divided by the total number of reflected echoes' - it seems like these are referring to 'own' echoes or first-order reflections, it is important to clarify this.

The reviewer is right. We have clarified it in the revised text, see lines 173-175.

Line 153: '..with a maximum difference of ...' - how is this difference calculated? What two quantities are being compared - not clear.

We have revised the text to clarify that the 14.3% value reflects the maximum difference in jamming probability between the RM and PK models, which occurred at a density of 10 bats. The values at each density are shown in Figure 2D, see lines 175-177.

Line 221: '..temporal aggregation helps..' - I'm assuming the authors meant temporal integration? However, I would caution against using the exact term 'temporal integration' as it is used in the field of audition to mean something different. Perhaps something like 'sensory integration', or 'multi-call integration'

To avoid ambiguity and better reflect the process modeled in our work, we have replaced the term "temporal aggregation" with "multi-call integration" throughout the revised manuscript. This term more accurately conveys the idea of combining information from multiple echolocation calls without conflicting with existing terminology.

(4) Discussion

Lines 302: 'Our model suggests...increasing the call-rate..' - not clear where this is explicitly tested or referred to in this manuscript. Can't see what was done to measure/quantify the effect of this variable in the Methods or anywhere else.

We have rephrased this paragraph as detailed in the Public Review above, see lines 346-349.

Line 319: 'spatial interference' - unclear what this means. This reviewer would strongly caution against creating new terms unless there is an absolute need for it. What is meant by 'interference' in this paper is hard to assess given that the word seems to be used as a synonym for jamming and also for actual physical wave-based interference.

We have rephrased this paragraph as detailed in the Public Review above, see line 119-120, 366-367.

Line 323: '..no benefit beyond a certain level...' - also not clear where this is explicitly tested. It seems like there was a set of simulations run for a variety of parameters but this is not written anywhere explicitly. What type of parameter search was done, was it all possible parameter combinations - or only a subset? This is not clear.

We have rephrased this paragraph as detailed in the Public Review above, see lines 372-375.

Line 324: '..ca. 110 dB-SPL.' - what reference distance?

All call levels were simulated and reported in dB-SPL, referenced at 0.1 meters from the emitting bat. We have clarified it in the revised text in the relevant contexts and specifically in line 529.

(5) Methods

Line 389: '...over a 2 x 1.5 m2 area..' It took a while to understand this statement and put it in context. Since there is no previous description of the entire L-arena, the reviewer took it to mean the simulations happened over the space of a 2 x 1.5 m2 area. Include a top-down description of the simulation's spatial setup and rephrase this sentence.

To address the confusion, we revised the text to clarify that the full simulation environment represents a corridor-shaped cave measuring 14.5×2.5 meters, with a right-angle turn located 5.5 meters before the exit, as shown in Figure 1A. The 2×1.5 m area refers specifically to the small zone at the far end of the cave where bats begin their flight. The

revised description now includes a clearer spatial overview to prevent ambiguity, see lines 456-460.

Line 398: Replace 'High proximity' with 'Close proximity'

Replaced.

Line 427: 'uniform target strength of -23 dB' - at what distance is this target strength defined? Given the reference distance can vary by echolocation convention (0.1 or 1 m), one can't assess if this is a reasonable value or not.

The reference distance for the reported target strength is 1 meter, in line with standard acoustic conventions. We have revised the text to clarify this explicitly (line 531).

Also, independent of the reference distance, particularly with reference to bats, the target strength is geometry-dependent, based on whether the wings are open or not. Using the entire wingspan of a bat to parametrise the target strength is an overestimate of the available reflective area. The effective reflective area is likely to be somewhere closer to the surface area of the body and a fraction of the wingspan together. This is important to note and/or mention explicitly since the value is not experimentally parametrised.

For comparison, experimentally based measurements used in Goetze et al. 2016 are -40 dB (presumably at 1 m since the source level is also defined at 1 m?), and Beleyur & Goerlitz 2019 show a range between -43 to -34 dB at 1 m.

We agree with the reviewer that target strength in bats is strongly influenced by their geometry, particularly wing posture during flight. In our model, we simplified this aspect by using a constant target strength, as the detailed temporal variation in body and wing geometry is pseudo-random and not explicitly modeled. We acknowledge that this is a simplification, and have now stated this limitation clearly in the revised manuscript. We chose a fixed value of -23 dB at 1 meter to reflect a plausible mid-range estimate, informed by anatomical data and consistent with values reported for similarly sized species (Beleyur and Goerlitz, 2019). To support this, we directly measured the target strength of a 3D-printed RM bat model, obtaining -32dB.

Moreover, a sensitivity analysis across a wide range (-49 to -23 dB) confirmed that performance metrics remain largely stable, indicating that our conclusions are not sensitive to this parameter, and suggesting that our results hold for different-sized bats. See lines 384-390, 533-538, and Supplementary Figures 3 and 4 in the revised article.

Line 434: 'To model the bat's cochlea...'. Bats have two cochleas. This model only describes one, while the agents are also endowed with the ability to detect sound direction - which requires two ears/cochleas.... There is missing information about the steps in between that needs to be provided.

We appreciate the reviewer's observation. Indeed, our model is monaural, and simulates detection using a single cochlear-like filter bank receiver. We have clarified this in the revised text to avoid confusion. This paragraph specifically describes the detection stage of the auditory processing pipeline. The localization process, which builds on detection and includes directional estimation, is described in the following paragraph (see line 583 onward), as discussed in the next comment and response.

Line 457: 'After detection, the bat estimates the range and Direction of Arrival...' This paragraph describes the overall idea, but not the implementation. What were the inputs and outputs for the range and DOA calculation performed by the agent? Or was this

information 'fed' in by the simulation framework? If there was no explicit DOA step that the agent performed, but it was assumed that agents can detect DOA, then this needs to be stated.

In the current simulation, the Direction of Arrival (DOA) was not modeled via an explicit binaural processing mechanism. Instead, based on experimental studies (Simmons et al., 1983; Popper and Fay, 1995), we assumed that bats can estimate the direction of an echo with an angular error that depends on the signal-to-noise ratio (SNR). Accordingly, the inputs to the DOA estimation were the peak level of the desired echo, noise level, and the level of acoustic interference. The output was an estimated direction of arrival that included a random angular error, drawn from a normal distribution whose standard deviation varied with the SNR. We have revised the relevant paragraph (Lines 583-592) to clarify this implementation.

Line 464: 'To evaluate the impact of the assumption...' - the 'self' and 'non-self' echoes can be distinguished perhaps using pragmatic time-delay cues, but also using spectro-temporal differences in individual calls/echoes. Do the agents have individual call structures, or do all the agents have the same call 'shape'? The echolocation parameters for the two modelled species are given, but whether there is call parameter variation implemented in the agents is not mentioned.

In our relatively simple model, all individuals emit the same type of chirp call, with parameters adapted only based on the distance to the nearest detected object. However, individual variation is introduced by assigning each bat a terminal frequency drawn from a normal distribution with a standard deviation of 1 kHz, as described in the revised version - lines 519-520. This small variation is not used explicitly as a spectro-temporal cue for echo discrimination.

In our model, all spectro-temporal variations—whether due to call structure or variations resulting from overlapping echoes from nearby reflectors—are processed through the filter bank, which compares the received echoes to the transmitted call during the detection stage. As such, the detection process itself can act as a discriminative filter, to some extent, based on similarity to the emitted call.

We acknowledge that real bats likely rely on a variety of spectro-temporal features for distinguishing self from non-self-echoes—such as call duration, received level, multi-harmonic structure, or amplitude modulation. In our simulation, we focus on comparing two limiting conditions: full recognition of self-generated echoes versus full confusion. Implementing a more nuanced self-recognition mechanism based on temporal or spectral cues would be a valuable extension for future work.

(6) References

Reference 22: Formatting error - and extra '4' in the reference.

The error has been fixed.

(7) Thoughts/comments

Even without 'recognition' of walls & conspecifics, bats may be able to avoid obstacles - this is a neat result. Also, using their framework the authors show that successful 'blind' object-agnostic obstacle avoidance can occur only when supported by some sort of memory. In some sense, this is a nice intermediate step showing the role of memory in bat navigation. We know that bats have good long-term and long-spatial scale memory,

and here the authors show that short-term spatial memory is important in situations where immediate sensory information is unreliable or unavailable.

We appreciate the reviewer's thoughtful summary. Indeed, one of the main takeaways of our study is that successful obstacle avoidance can occur even without explicit recognition of walls or conspecifics—provided that a clustered multi-call integration is in place. Our model shows that when immediate sensory information is unreliable, integrating detections over time becomes essential for effective navigation. This supports the broader view that memory, even on short timescales, plays an important role in bat behavior.

(8) Reporting GLM results

The p-value, t-statistic, and degrees of freedom are reported consistently across multiple GLM results. However, the most important part which is the effect size is not consistently reported - and this needs to be included in all results, and even in the table. The effect size provides an indicator of the parameter's magnitude, and thus scientific context.

We agree that the effect size provides essential scientific context. In fact, we already include the effect size explicitly in Table 1, as shown in the “Effect Size” column for each tested parameter. These values describe the magnitude of each parameter's effect on exit probability, jamming probability, and collision rate. In the main text, effect sizes are presented as concrete changes in performance metrics (e.g., “exit probability increased from 20% to 87%,” or “with a decrease of $3.5\% \pm 8\%$ to $5.5\% \pm 5\%$ (mean $\pm$ s.e.)”), which we believe improves interpretability and scientific relevance.

To further clarify this in the main text, we have reviewed the reported results and ensured that effect sizes are mentioned more consistently wherever GLM outcomes are discussed. Additionally, we have added a brief note in the table caption to emphasize that effect sizes are provided for all tested parameters.

The 'tStat' appears multiple times and seems to be the output of the MATLAB GLM function. This acronym is specific to the MATLAB implementation and needs to be replaced with a conventionally used acronym such as 't', or the full form 't-statistic' too. This step is to keep the results independent of the programming language used.

We have replaced all instances of tStat with the more conventional term ‘t’ throughout the manuscript to maintain consistency with standard reporting practices.

Reviewer #2 (Recommendations for the authors):

In addition to my public review, I had a few minor points that the authors may want to consider when revising their paper.

(1) Figures 2, 3, and 4 may benefit from using different marker styles, in addition to different colors, to show the different cases.

Thank you for the suggestion. In Figures 2–4, the markers represent means with standard error bars. To maintain clarity and consistency across all conditions, we have chosen to keep a standardized marker style – and we clarify this in the legend. We found that varying only the colors is sufficient for distinguishing between conditions without introducing visual clutter.

(2) The text "PK" in the inset for Figure 2A is very difficult to read. I would suggest using grey as with "RM" in the other inset.

We have updated the insert in Figure 2A to improve legibility.

(3) Are the error bars in Figure 3 very small? I wasn't able to see them. If that is the case, the authors may want to mention this in the caption.

You are correct—the error bars are present in all plots but appear very small due to the large number of simulation repetitions and low variability. We have revised the caption to explicitly mention this.

(4) The species name of PK is spelled inconsistently (kuhli, khulli, and kuhlii).

We have corrected the species name throughout the manuscript.

(5) Table 1 is a great condensation of all the results, but the time to exit is missing. It may be helpful if summary statistics on that were here as well.

We have added time-to-exit to the effect size column in Table 1, alongside the other performance metrics, to provide a more complete summary of the simulation results.

(6) I may have missed it, but why are there two values for the exit probability when nominal flight speed is varied?

The exit probability was not monotonic with flight speed, but rather showed a parabolic trend with a clear optimum. Therefore, we reported two values representing the effect before and after the peak. We have clarified this in the revised table and updated the caption accordingly.

(7) Table 2 has an extra header after the page break on page 18.

The extra header in Table 2 after the page break has been removed in the revised manuscript.

(8) The G functions have 2 arguments in their definitions and Equation 1, but only one argument in Equations 2 and 3. I wasn't able to see why.

Thank you for pointing this out. You are correct—this was a typographical error. We have corrected the argument notation in Equations 2 and 3 and explicitly included the frequency dependence of the gain (G) functions in both equations.

(9) D_{txrx} was not defined but it was used in Equation 2.

The variable D_{txrx} is defined in the equation notation section as: $D_{(tx)r_x}$ – the distance [m] between the transmitting conspecific and the receiving focal bat, from the transmitter's perspective. We have now ensured that this definition is clearly linked to Equation 2 in the revised text. Moreover, we have added a supplementary figure that illustrates the geometric configuration defined by the equations to further support clarity, as described in the Public Review above.

(10) It was hard for me to understand what was meant by ϕ_{rx} and ϕ_{tx} . These were described as angles between the rx or tx bats and the target, but I couldn't tell what the point defining the angle was. Perhaps a diagram would help, or more precise definitions.

We have revised the caption to provide clearer and more precise definitions. Additionally, we have included a geometric diagram as a supplementary figure, as noted in the Public Review above, to visually clarify the spatial relationships and angle definitions used in the equations, see lines 498-499.

(11) *Was the hearing threshold the same for both species?*

Yes. We have clarified it in the revised version.

(12) *Collision avoidance is described as turning to the "opposite direction" in the supplemental figure explaining the model. Is this 90 degrees or 180 degrees? If 90 degrees, how do these turns decide between right and left?*

In our model, the bat does not perform a fixed 90° or 180° turn. Instead, the avoidance behavior is implemented by setting the maximum angular velocity in the direction opposite to the detected echo. For example, if the obstacle or conspecific is detected on the bat's right side, the bat begins turning left, and vice versa.

This turning direction is re-evaluated at each decision step, which occurs after every echolocation pulse. The bat continues turning in the same direction if the obstacle remains in front, otherwise it resumes regular pathfinding. We have clarified this behavior in the updated figure caption and model description, see lines 478-493.

Reviewer #3 (Recommendations for the authors):

(1) *Lines 27-31: These sentences mischaracterize the results. This claim appears to equate "the model works" with "this is what bats actually do." Also, the model does not indicate that bats' echolocation strategies are robust enough to mitigate the effects of jamming - this is self-evident from the fact that bats navigate successfully via echolocation in dense groups.*

Thank you for the comment. Our aim was not to claim that the model confirms actual bat behavior, but rather to demonstrate that simple and biologically plausible strategies—such as signal redundancy and basic pathfinding—are sufficient to explain how bats might cope with acoustic interference in dense settings. We have revised the wording to better reflect this goal and to avoid overinterpreting the model's implications.

See abstract in the revised version.

(2) *Line 37: This number underestimates the number of bats that form some of the largest aggregations of individuals worldwide - the free-tailed bats can form aggregations exceeding several million bats.*

We have revised the text to reflect that some bat species, such as free-tailed bats, are known to form colonies of several million individuals, which exceed the typical range. The updated sentence accounts for these extreme cases, see lines 36-37.

(3) *The flight densities explained in the introduction and chosen references are not representative of the literature - without providing additional justification for the chosen species, it can be interpreted that the selection of the species for the simulation is somewhat arbitrary. If the goal is to model dense emergence flight, why not use a species that has been studied in terms of acoustic and flight behavior during dense emergence flights---such as Tadarida brasiliensis?*

Our goal was to develop a general model applicable to a broad class of FMeolocating bat species. The two species we selected—*Pipistrellus kuhlii* (PK) and *Rhinopoma microphyllum* (RM)—span a wide range of signal characteristics: from wideband (PK) to narrowband (RM), providing a representative contrast in call structure.

Although we did not include *Tadarida brasiliensis* (TB) specifically, its echolocation calls are acoustically similar to RM in terminal frequency and fall between PK and RM in bandwidth. Therefore, we believe our findings are likely to generalize to TB and other FM-bats.

Moreover, as noted in a previous response, the average inter-bat distance in our highest-density simulations (0.27 m) is still smaller than those reported for *Tadarida brasiliensis* during dense emergences—further supporting the relevance of our model to such scenarios.

To support broader applicability, we also provide a supplementary graphical user interface (GUI) that allows users to modify key echolocation parameters and explore their impact on behavior—making the framework adaptable to additional species, including TB.

(4) Line 78: It is not clear how (or even if) the simulated bats estimate the direction of obstacles. The explanation given in lines 457-463 is quite confusing. What is the acoustic/neurological mechanism that enables this direction estimation? If there is some mechanism (such as binaural processing), how does this extrapolate to 3D?

This comment echoes a similar concern raised by a previous reviewer. As explained earlier, in the current simulation, the Direction of Arrival (DOA) was not modeled via an explicit binaural processing mechanism. The complete is detailed in to Reviewer #1, Line 457. This implementation is now clarified in the revised text, and a detailed description of the localization process is also provided in the Methods section (lines 583-592).

(5) The authors propose they are modeling the dynamic echolocation of bats in the simulation (line 79), but it appears (whether this is due to a lack of information in the manuscript or true lack in the simulation) that the authors only modeled a flight response. How did the authors account for bats dynamically changing their echolocation? This is unclear and from what I can tell may just mean that the bats can switch between foraging phase call types depending on the distance to a detected obstacle. Can the authors elaborate more on this?

The echolocation behavior of the bats—including dynamic call adjustments— was implemented in the simulation and is described in detail in the Methods section (lines 498-520 and Table 2). To avoid redundancy, the Results chapter originally referred to this section, but we have now added a brief explanation in the Results to clarify that the bats' call parameters (IPI, duration, and frequency range) adapt based on the distance to detected objects, following empirically documented echolocation phases ("search," "approach," "buzz"). These dynamics are consistent with established bat behavior during navigation in cluttered environments such as caves.

(6) Figure 1 C3: "Detection threshold": what is this and how was it derived?

The caption also mentions yellow arrows, but they are absent from the figure. C4: Each threshold excursion is marked with an asterisk, but there are many more excursions than asterisks. Why are only some marked? Unclear.

C3: The detection threshold is determined dynamically. It is set to the greater of either 7 dB above the noise level (0 dB-SPL)(Kick, 1982; Saillant *et al.*, 1993; Sanderson *et al.*, 2003; Boonman *et al.*, 2013) or the maximal received level minus 70 dB, effectively applying a

dynamic range of 70 dB. This clarification has been added to the Methods section. The yellow arrow has been added.

C4: Thank you for this important observation. Only peaks marked with asterisks represent successful detections—those that were identified in both the interference-free and full detection conditions, as explained in the Methods. Other visible peaks result from masking signals or overlapping echoes from nearby reflectors, but they do not meet the detection criteria. To keep the figure caption concise, we have elaborated on this process more clearly in the revised Methods section. We added this information to the legend

| (7) Figure 2: A line indicating RM, No Masking is absent

Thank you for pointing this out. The missing line for RM, No Masking has now been added in the revised version of Figure 2.

| (8) Line 121: "reflected off conspecifics". Does this mean echoes due to conspecifics?

The phrase "reflected off conspecifics" refers to echoes originating from the bat's own call and reflected off the bodies of nearby conspecifics. We have clarified the wording in the revised text to avoid confusion

| (9) Line 125: Why are low-frequency channels stimulated by higher frequencies? This needs further clarification.

The cochlear filter bank in our model is implemented using gammatone filters, each modeled as an 8th-order Butterworth filter. Due to the non-ideal filter response and relatively broad bandwidths—especially in the lower-frequency channels—strong energy from the beginning of the downward FM chirp (at higher frequencies) can still produce residual activation in lower-frequency channels. While these stimulations are usually below the detection threshold, they may still be visible as early sub-threshold responses. Given the technical nature of this explanation (a property of the filter implementation) and it does not influence the detection outcomes, we have chosen not to elaborate on it in the figure caption or Methods.

| (10) Lines 146-150: This is an interesting finding. Is there a theoretical justification for it?

This outcome arises directly from the simulation results. As noted in the Discussion (lines 359-365), although *Pipistrellus kuhlii* (PK) shows a modest advantage in jamming resistance due to its broader bandwidth, the redundancy in sensory information across calls—enabled by frequent echolocation—appears to compensate for these signal differences. As a result, the small variations in echo quality between species do not translate into significant differences in performance. We speculate that if the difference in jamming probability had been larger, performance disparities would likely have emerged.

| (11) Line 151: The authors define a jammed echo as an echo entirely missed due to masking. Is this appropriate? Doesn't echo mis-assignment also constitute jamming?

We agree that echo mis-assignment can also degrade performance; however, in our model, we distinguish between two outcomes: (1) complete masking (echo not detected), and (2) detection with a localization error. As explained in the Methods (lines 500–507), we run the detection analysis twice—once with only desired echoes ("interference-free detection") and once including masking signals ("full detection"). If a previously detected echo is no longer detected, it is classified as a jammed echo. If the echo is still detected but the delay shifts by more than 100 μ s compared to the interference-free condition, it is also considered jammed.

If the delay shift is smaller, it is treated as a detection with localization error rather than full jamming. We have clarified this distinction in the revised Methods section.

(12) *Figure 2-E: Detection probability statistics are of limited usefulness without accompanying false alarm rate (FAR) statistics. Do the authors have FAR numbers?*

We understand FAR to refer to instances where masking signals or other acoustic phenomena are mistakenly interpreted as real echoes from physical objects. As explained in the manuscript, we implemented two model versions: one without confusion, and one with full confusion.

Figure 2E reports detection performance under the non-confusion model, in which only echoes from actual physical reflectors are used, and no false detections occur—hence, the false alarm rate is effectively zero in this condition. In the full-confusion model, all detected echoes—including those originating from masking signals or conspecific calls—are treated as valid detections, which may include false alarms. However, we did not explicitly quantify the false alarm rate as a separate metric in this simulation.

We agree that tracking FAR could be informative and will consider incorporating it into future versions of the model.

(13) *Line 161: RM bats suffered from a significantly higher probability of the "desired conspecific's echoes" being jammed. What does "desired conspecific's echoes" mean? This is unclear.*

The term “desired conspecific's echoes” refers to echoes originating from the bat’s own call, reflected off nearby conspecifics, which are treated as relevant reflectors for collision avoidance. We have revised the wording in the text for clarity.

(14) *Line 188: Why didn't the size of the integration window affect jamming probability? I couldn't find this explained in the discussion.*

The jamming probability in our analysis is computed at the individual-echo level, prior to any temporal integration. Since the integration window is applied *after* the detection step, it does not influence whether a specific echo is masked (i.e., jammed) or not. Therefore, as expected, we did not observe a significant effect of integration window size on jamming probability.

(15) *Line 217-218: Why do the authors think this would be?*

Thank you for the thoughtful question. We agree that, in theory, increasing call intensity should raise the levels of both desired echoes and masking signals proportionally. However, in our model, the environmental noise floor and detection threshold remain constant, meaning that higher call intensities increase the signal-to-noise ratio (SNR) more effectively for weaker echoes, especially those at longer distances or with low reflectivity. This could lead to a higher likelihood of those echoes crossing the detection threshold, resulting in a small but measurable reduction in jamming probability.

Additionally, the non-linear behavior of the filter-bank receiver—including such as thresholding at multiple stages—can introduce asymmetries in how increased signal levels affect the detection of target versus masking signals.

That said, the effect size was small, and the improvement in jamming probability did not translate into any significant gain in behavioral performance (e.g., exit probability or collision rate), as shown in Figure 3C.

(16) Line 233: I'm not sure I understand how a slightly improved aggregation model that clustered detected reflectors over one-second periods is different. Doesn't this just lead to on average more calls integrated into memory?

While increasing the memory duration does lead to more detections being available, the enhanced aggregation model (we now refer to as multi-call clustering) differs fundamentally from the simpler one. As detailed in the Methods, it includes additional processing steps: clustering spatially close detections, removing outliers, and estimating wall directions based on the spatial structure of clustered echoes. In contrast, the simpler model treats each detection as an isolated point without estimating obstacle orientation. These additional steps allow for more robust environmental interpretation and significantly improve performance under high-confusion conditions. We have clarified it in revised text (lines 606-616) and added a Supplementary Figure 2B.

(17) Table 1: What about conspecific target strength?

We have now added the conspecific target strength as a tested parameter in Table 1, along with its tested range, default value, and measured effect sizes. A detailed sensitivity analysis is also presented in Supplementary Figure 4, demonstrating that variations in conspecific target strength had relatively minor effects on performance metrics.

(18) Figure 3-A: The x-axis is the number of calls in the integration window. But the leftmost sample on each curve is at 0 calls. Shouldn't this be 1?

“0 calls” refers to the case where only the most recent call is used for pathfinding—without integrating any information from prior calls. The x-axis reflects the number of previous calls stored in memory, so a value of 0 still includes the current call. We’ve clarified this terminology in the figure caption.

(19) Lines 282-283: This statement needs to be clarified that it is with the constraints of using a 2D simulation with at most 33 bats/m². It also should be clarified that it is assumed the bat can reliably distinguish between its own echoes and conspecific echoes, which is a very important caveat.

We have revised the text to clarify that the results are based on a 2D simulation with a maximum tested density of 33 bats/m². We also now explicitly state that the model assumes bats can distinguish between their own echoes and those generated by conspecifics—an assumption we recognize as a simplification. These clarifications help place the results within the scope and constraints of the simulation. Moreover, as described in the text (and noted in previous response): the average distance to the nearest bat in our simulation is 0.27m (with 100 bats), whereas reported distances in very dense colonies are 0.5m

(20) Line 294: What is this sentence referring to?

The sentence refers to the finding that, even under high bat densities, a substantial portion of the echoes—particularly those reflected from nearby obstacles (e.g., 1 m away)—were jammed due to masking. Nevertheless, the bats in the simulation were still able to navigate successfully using partial sensory input. We have clarified the sentence in the revised text to make this point more explicit, see line 333-336.

(21) Line 302: Was jamming less likely when IPI was higher or lower? I could not find this demonstrated anywhere in the manuscript.

We agree that the original text was not sufficiently clear on this point. While we did not explicitly test fixed IPI values as a parameter, the model does simulate the natural behavior of decreasing IPI as bats approach obstacles. This behavior is supported by empirical observations and is incorporated into the echolocation dynamics of the simulation. We have clarified this point in the revised text (see Lines 346-351) and explained that while lower IPI introduces more acoustic overlap, it also increases redundancy and improves detection through temporal integration.

(22) Lines 313-314: This is an interesting assumption, but it is not evident that is substantiated by the references.

The claim is based on well-established principles in signal processing and bioacoustics. Wideband signals—such as those emitted by PK bats— distribute their energy over a broader frequency range, which makes them inherently more resistant to narrowband interference and masking. This concept is commonly applied in both biological and artificial sonar systems and is supported by empirical studies in bats and theory in acoustic sensing.

For example, Beleyur & Goerlitz (2019) demonstrate that broader bandwidth calls improve detection in cluttered and jamming-prone environments. Similarly, Ulanovsky et al. (2004) and Schnitzler & Kalko (200) discuss how FM bats' wideband calls enhance temporal and spatial resolution, helping to reduce the impact of overlapping signals from conspecifics. These findings align with communication theory where spread-spectrum techniques improve robustness in noisy environments.

We agree with the reviewer that this is an important point and we have updated the manuscript to clarify this rationale and cite the relevant literature accordingly – lines 631-363,

(23) Lines 318-319: What is the justification for "probably"? Isn't this just a supposition?

We agree with the reviewer's point and have rephrased the sentence

(24) Line 320: How does this 63% performance match the sentence in line 295?

The sentence in Line 295 refers to the overall ability of the bats to navigate successfully despite high jamming levels, highlighting the robustness of the strategy under challenging conditions. The figure in Line 320 (63%) quantifies this performance under the most extreme simulated scenario (100 bats / 3 m²), where both spatial and acoustic interferences are maximal. We have rephrased the text in the revised version (lines 324-327).

(25) Lines 341-345: It seems like this is more likely to be the main takeaway of the paper.

As noted in the Public Review above, there is substantial literature supporting the assumption that bats can recognize their own echoes and distinguish them from those of conspecifics (e.g., Schnitzler, Bioscience, 2001; Kozlowski et al., 2001, 2008; Burnett & Masters, 2002; Chiu et al., 2009; Yovel et al., 2009; Beetz & Hechavarría, 2022). Therefore, we consider our assumption of self-recognition to be well-supported, at least under typical conditions. That said, we agree that the impact of echo confusion on performance is significant and highlights a critical challenge in dense environments.

To our knowledge, this is the first computational model to explicitly simulate both self-recognition and full echo confusion under high-density conditions. We believe that the combination of modeled constraints and the demonstrated robustness of simple

sensorimotor strategies, even under worst-case assumptions, is what makes this contribution both novel and meaningful.

| (26) Lines 349-350: What is the aggregation model? What is meant by "integration"?

We have revised the text to clarify that the “aggregation model” refers to a multi-call clustering process that includes clustering of detections, removal of outliers, and estimation of wall orientation, as described in detail in the revised Methods and Results sections.

| (27) Line 354: Again, why isn't this the assumption we're working under?

As addressed in our response to Comment 25, our primary model assumes that bats can recognize their own echoes—an assumption supported by substantial empirical evidence. The alternative “full confusion” model was included to explore a worst-case scenario and highlight the behavioral consequences of failing to distinguish self from conspecific echoes. We assume that real bats may experience some degree of echo misidentification; however, our assumption of full confusion represents a worst-case scenario.

| (28) Line 382: "Under the assumption that..." I agree that bats probably can, but if we assume they can differentiate them all, where's the jamming problem?

The assumption that bats can *theoretically* distinguish between different signal sources applies after successful detection. However, the jamming problem arises during the detection and localization stages, where acoustic interference can prevent echoes from crossing the detection threshold or distort their timing.

| (29) Lines 386-387: The paper referenced focused on JAR in the context of foraging. What changes were made to the simulation to switch to obstacle avoidance?

While the simulation framework in Mazar & Yovel (2020) was developed to study jamming avoidance during foraging, the core components—such as the acoustic calculations, receiver model, and echolocation behavior—remain applicable. For the current study, we adapted the simulation extensively to address colony-exit behavior. These modifications include modeling cave walls as acoustic reflectors, implementing a pathfinding algorithm, integrating obstacle-avoidance maneuvers, and adapting the integration window and integration processes. These updates are detailed throughout the Methods section.

| (30) Line 400-402: Something doesn't add up with the statement: each decision relies on an integration window that records estimated locations of detected reflectors from the last five echolocation calls, with the parameter being tested between 1 and 10 calls. Can the authors reword this to make it less confusing?

We have reworded the sentence to clarify that the default integration window includes five calls, while we systematically tested the effect of using 1 to 10 calls, see lines 486-487.

| (31) Line 393: "30 deg/sec" why was this value chosen?

The turning rate of 30 deg/sec was manually selected to approximate the curvature of natural foraging flight paths observed in *Rhinopoma microphyllum* using on-board tags. Moreover, in Mazar & Yovel (2020), we showed that the flight dynamics of simulated bats in a closed room closely matched those of *Pipistrellus kuhlii* flying in a room of similar dimensions. However, in the current simulation, bats rarely follow a random-walk trajectory due to the structured environment and frequent obstacle detection. As a result, this parameter has no meaningful impact on the simulation outcomes.

(32) Line 412: "Harmony" --- do you mean harmonic? And what is the empirical evidence that RM bats use the 2nd harmonic compared to the 1st?

Perhaps showing a spectrogram of a real RM signal would be helpful.

The typo-error was corrected. For reference See (Goldshtein et al., 2025)

(33) Table 2: Something is incorrect with the table. The first row on the next page is the wrong species name. Also, where are the citations for these parameter values?

The table header has been corrected in the revised version. The parameter values for flight and echolocation behavior were derived from existing literature and empirical data: *Pipistrellus kuhlii* parameters were based on Kalko (1995), and *Rhinopoma microphyllum* parameters were extracted from our own recordings using on-board tags, as described in Goldstein et al. (2025). We have added the appropriate citations to Table 2.

(34) Line 442: How was the threshold level chosen?

The detection threshold in each level is set to the greater of either 7 dB above the noise level (0 dB-SPL) or the maximal received level minus 70 dB, effectively applying a dynamic range of 70 dB.

(35) Line 445: 100 micros: This is about 3cm. The resolution of PK is about 1cm. For RM it's about 10cm. So, this window is generous for PK, but too strict for RM.

To keep the model simple and avoid introducing species-specific detection thresholds, we selected a biologically plausible compromise that could reasonably apply to both species. This simplification ensures consistency across simulations while remaining within the known behavioral range.

(36) Line 448: What is the spectrum of the Gaussian noise, and did it change between PK and RM?

We used the same white Gaussian noise with a flat spectrum across the relevant frequency range (10–80 kHz) for both species. We have clarified this in the revised text in lines 570-572.

(37) Line 451: 4 milliseconds is 1.3m. Is this appropriate?

The 4 milliseconds window was selected based on established auditory masking thresholds described in Mazar & Yovel (2020), and supported by (Popper and Fay, 1995) ch. 2.4.5, ((Blauert, 1997), ch. 3.1 and (Mohl and Surlykke, 1989). These values provide conservative lower bounds on bats' ability to cope with masking (Beleyur and Goerlitz, 2019). For simplicity, we used constant thresholds within each window, see lines 574-576.

(38) Line 452: Citation for the forward and backward masking durations?

See the to the previous comment.

(39) Lines 460-461: This is unclear. How does the bat get directional information? The authors claim to be able to measure direction-of-arrival for each detection, but it is not clear how this is done

As noted in our response to Reviewer 1 (Comment on Line 457), directional information is not computed via an explicit binaural model. Instead, we assume the bat estimates the direction of arrival with an angular error that depends on the SNR, based on established studies (e.g., Simmons et al., 1983; Popper & Fay, 1995). We have clarified this in the revised text in lines 583-592.

(40) Line 467: It seems like the authors are modeling pulse-echo ambiguity, at least in this one alternative model, which is good! However the alternative model doesn't get much attention in the paper. Is there a reason for this?

We would like to clarify that we did not model pulse-echo. In our confusion model, all echoes received within the IPI are attributed to the bat's most recent call. This includes echoes that may in fact originate from conspecific calls, but the model does not assign self-echoes to earlier pulses or span multiple IPIs. Therefore, while the model captures echo confusion, it does not include true pulse-echo ambiguity. We have clarified this point in the revised text in lines 551-553.

(41) Line 41: "continuous" is more appropriate than "constant".

Thank you, we have rephrased the text accordingly.

(42) Line 69: "band width" should be one word.

Thank you, we have corrected it to "bandwidth".

(43) Line 79: "bats" should be in the possessive.

Thank you, the text has been rephrased.

(44) Line 128: "convoluted" don't you mean "convolved"?

We have replaced "convoluted" with the correct term "convolved" in the revised text.

(45) Please check your references, as there are some incomplete citations and typos.

Thank you, we have reviewed and corrected all references for completeness and consistency.

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